Where the Bits Must Live: A Tight Input-Side Information Bound for Fixed-State Spectral Models

Anonymous submission

Abstract

We prove a tight lower bound on the information that must be supplied on the INPUT SIDE of a fixed-state attention-free sequence model: exact H-hop content retrieval requires at least log2|A| bits of key/value role information, where |A| is the number of distinct answers compatible with the visible symbol sequence (Theorem A). The difficulty is that the bound is universal -- it holds for ANY deterministic decoder, independent of depth, state order and kernel -- so the missing bits cannot be manufactured inside the model; placing them in the kernel makes its coefficients time-varying, which by Toeplitz IFF time-invariance destroys the single-FFT / IIR duality. We therefore characterize WHERE the bits may live: an explicit tag, an alphabet partition, and a positional format prior each carry one bit per ambiguous token, but only the alphabet partition preserves duality while remaining free of run-time annotation. Strongest evidence: exhaustive enumeration gives accuracy 2^(B-m) for B<m and exactly 1 at B=m for m=1..4; a real static-LTI stack of depth 1..8 produces bit-identical outputs across all compatible role assignments (deviation 0.0e+00); the three carriers measure 1.0000 / 1.0000 / 0.2969. Significance: the objection 'the input format solves the task' becomes a quantitative design rule -- budget log2|A| bits at the input or relax exactness, because extra layers cannot substitute. This REFINES our own earlier 'exactly one bit' claim into a general total-bit bound. Two tight structural bounds (an order-2 state for one hop; exactly H static-LTI layers for H hops under last-layer read-out) are attained by the same Toeplitz, O(1) construction. We make no language-modeling, perplexity or throughput claim.

Table 1 (first page). The most common rejection reason for this line of work is 'isn't this known in the SSM / linear-attention literature?'. The right column states the exact increment, with sharp language (IFF, tight bound) rather than 'we study' or 'we improve'.

Known (with source)	This paper: exact increment
LTI duality requires time-invariant coefficients (S4 [1]; S5 [2]; Gu et al., 2022)	T4: Toeplitz IFF parameter-time-invariant for ARBITRARY order and per-channel grids; c_t != 0 is an essential premise, not a technicality.
Fixed-state models are weak at associative recall (Zoology [13]; [14]; [15])	T5: impossibility for ANY read-out (including nonlinear) under a fixed linear history summary; scope stated as four explicit premises.
Selective/input-dependent parameters break the convolution view (Mamba [4])	T2 construction keeps coefficients static and still attains multi-hop content addressing; duality error 2.27e-16.
Linear attention uses S = sum k_s v_s^T ([8]; GLA [6])	T6: at rho=1, single pole the state is isomorphic; rho<1 multi-pole K(d)=sum_p w_p rho_p^d is a decay-spectrum weighted operator. We do NOT claim to leave the family.
Outer-product / bilinear read-out can implement binding (TTT, arXiv:2407.04620)	T1+T2 JOINT tightness: minimum tensor order 2 AND exactly H layers, with O(1) state, static Toeplitz kernel and no N x N, all simultaneously.
Fixed-state models are weak at recall; input format can carry task information (Zoology [13]; [14]; [15])	THEOREM A (main increment): a TIGHT log2|A|-bit lower bound on INPUT-SIDE role information; a proof that Toeplitz duality FORCES those bits to the input; three-way carrier conservation; a RIGHT-ANGLE bit-depth frontier (no depth can substitute).

1 Introduction

A fixed-size state must summarize an unbounded history. The literature has established the LTI convolution/recurrence duality, the time-invariance requirement for a convolution view, and the recall/copying weakness of fixed-state models (S4 [1], S5 [2], Zoology [13], Repeat After Me [14], Illusion of State [15]). We do NOT re-prove those. The question we answer is which ALGEBRAIC property of a specific architecture produces each boundary, and which single premise must be removed to escape it.

One story. Exact retrieval needs information that a flat symbol sequence does not contain: which token of each binding pair is the key and which is the value. We prove a tight lower bound on that information (Theorem A), prove that a Toeplitz kernel cannot create it internally, and show the bit-depth frontier is a right angle. Two tight structural bounds complete the picture: an order-2 state for one hop (Theorem 1) and exactly H static-LTI layers for H hops under last-layer read-out (Theorem 2). All three are attained by one Toeplitz, O(1)-state construction. Supporting statements and withdrawals are kept in the appendix as a soundness record, not as contributions.

1.2 Contributions

· **1. A tight input-side information bound (Theorem A, main result).** Exact retrieval needs B >= log2|A| bits of role information; the bound is universal, attained, and independent of depth. Within the static-LTI class the bits are FORCED to the input by Toeplitz duality. Parts (a),(b) are standard information theory -- we say so explicitly; the novelty is the location theorem, three-way conservation and the right-angle frontier.

· **2. Two tight structural bounds.** Minimum tensor order exactly 2 for one hop (T1), and exactly H static-LTI layers for H hops (T2). Both are attained by one construction with O(1) state and no N x N scores.

· **3. A constructive counterexample.** An orthogonal codebook partition supplies the required information without any run-time annotation and keeps FFT/IIR duality (relative error 2.89e-16). This retires our earlier mutual-exclusivity claim and the positional-format prior.

· **4. Falsifiable witnesses.** Exhaustive bit enumeration (m=1..4), depth-1..8 invariance of a real static-LTI stack, three carrier measurements, and a 5 x 5 layer x hop grid. Every number is injected from raw result files; no value is a theorem.

We deliberately present FOUR core results, not thirty. Honesty about withdrawals is a soundness practice, not an originality claim.

2 Preliminaries and Notation

We use **t** for the absolute time index, **d = t - s** for a delay, **tau** for a kernel lag, **H** for the number of hops, **H_L** for the number of layers, and **n** for the number of pairs. These are kept distinct throughout; no symbol is reused.

Notation ledger (v6, final). C in R^(V x V) is the deterministic Sylvester Hadamard code, C C^T = I, NOT learned, applied by FWHT. V = 2 V_K; the first V_K rows are KEY codes, the last V_K are VALUE codes -- this is the SINGLE role bit. K_h(d) = sum_p w_hp rho_hp^d > 0 is static and time-invariant. A_t = C[x_(t-1)] and b_t = C[x_t] are kernel-external; there is NO r[t] gate (the parity mask was WITHDRAWN, the withdrawn positional format prior). q_(h+1) = normalize(T_h z_h) with T_h a FIXED learned linear map, not a function of t or of the content. s_(h,p) are P state vectors per layer; total state is O(H_L P V) scalars with NO dependence on L.

A content-chain task supplies n pairs (k_p, k_(p+1)) with DISTINCT keys; each pair is emitted as two ADJACENT symbols (KEY then VALUE), pair ORDER is arbitrary, and any number of zero FILLER tokens may appear between pairs. The query q_1 = C[k_0] is supplied EXTERNALLY; the answer is k_H; chance is 1/V_K. Nothing in the input marks a pair boundary or a symbol role at run time. Section 3 states every premise on the same page as its theorem; Sections 3.4-3.6 give the COMPLETE proofs in the main text.

2.1 The TSC-H layer: formal specification and state accounting

symbol	meaning	depends on
V, V_K	codebook rows; V = 2 V_K, C C^T = I	nothing
C[k]	KEY code for id k (row k)	id only
C[V_K+k]	VALUE code for id k (row V_K+k)	id only
x_t	symbol id at t; 0 for filler	data
A_t = C[x_(t-1)]	predecessor signal	data
b_t = C[x_t]	current signal	data
K_h(d)	sum_p w_hp rho_hp^d, 0 < rho < 1	d only (STATIC)
S_h	sum_t K_h(d) A_t (x) b_t, order-2 state	prefix <= L-1
q_h	layer query; q_1 = C[k_0]	external query
T_h	U_h W_h^T/sqrt(r), r >= n+1 (R4)	learned parameter
z_h = S_h^T q_h	sum_t K_h(d) <q_h,A_t> b_t	state + query
tau	output temperature, argmax-invariant	1 scalar

Layer recurrence (all operators static). For h = 1..H_L: m_h[t] = <q_h, A_t>; z_h = sum_t K_h(L-1-t) m_h[t] b_t; and q_(h+1) = normalize(T_h z_h). The read-out is logits_j = tau <z_(H_L), C[V_K+j]>. Because T_h is a FIXED parameter (it does not vary with t and does not depend on the content), K_h depends only on the delay d = L-1-t and the impulse-response matrix is Toeplitz. Hence a single FFT realizes exactly the same operator as the P-tap IIR scan (Theorem 1, Section 3.4).

Exact state accounting. Each layer stores P vectors of length V plus one query vector; with P=2 and fp32 the measured bound is 512 bytes per layer for V=64, INDEPENDENT of L (verified at L = 9, 65, 1025, 4097). The total is linear in H_L and constant in L. There is no N x N score matrix, no softmax over tokens, and no KV cache. Parameter count is (H_L-1) x 2 V r + 1; the codebook itself stores ZERO bytes because C is applied by FWHT. At V = 2^18 with r = 8 this is 4194304 parameters, versus 549.8 GB for a dense codebook (measured FWHT 0.0119 s per vector).

3 Four Core Results

We present ONE main theorem (A), two tight structural bounds (T1, T2) and one constructive counterexample. Standard facts are demoted to propositions in the appendix; the full engineering ledger is NOT part of the submission's contribution. Every theorem states its premises in the same section, with no cross-reference required.

3.1 T1 - Minimum tensor order is exactly 2

Premises (complete). A single static-LTI mixer with a fixed-dimensional state; the query enters only at the final position; values are independently assignable; at least two distinguishable targets; the read-out may be ANY function of the state and the query.

Claim. (a) An order-1 vector state cannot implement exact retrieval for any read-out. (b) An order-2 tensor state attains it exactly with zero interference and without knowing the target position. Therefore the minimum order is 2 and the bound is TIGHT. **Proof route (corrected).** The summary is A = sum_p [alpha_p k_p + beta_p v_p] with beta_p = K(L-s_p) > 0 for a strictly positive static kernel. A nonzero kernel of the induced map L: R^(nd) -> R^d does NOT by itself guarantee that a kernel vector perturbs the TARGET coordinate. We construct one explicitly: for n>=2 take Delta v_1 = beta_2 u, Delta v_2 = -beta_1 u and all other blocks zero, with any nonzero u. Then sum_p beta_p Delta v_p = 0, so Delta lies in ker L, yet it changes the first value block. Choosing the query to target that block gives two value assignments with identical (A,q) but different required answers, so NO read-out can be correct. (b) The outer-product state preserves the binding. Full proof: Appendix A.1/A.13.

Scope. Corollary B1.1 ('pairwise matching is necessary') is an EXPLANATORY FRAMEWORK, NOT a theorem; it does not prove that pairwise matching is the only possible escape. TSC escapes by changing the tensor order of the SIGNAL, not by introducing a pairwise score.

3.2 T2 - Exactly H static-LTI layers (new)

Premises (complete). H1: n pairs (k_p, k_(p+1)), distinct keys, strict key/value alternation, arbitrary pair-position permutation. H2: query k_0 at t=L-1. H3: fixed code C with C C^T = I on the key range. H4: H_L layers, each with static kernel K_h(d)=sum_p w_hp rho_hp^d > 0. H5 (REVISED by Theorem A): the INPUT ENCODING supplies at least log2|A| bits of key/value role information, where |A| is the number of answers compatible with the visible symbol sequence. The orthogonal codebook partition (KEY symbol k uses C[k], VALUE symbol k uses C[V_K+k], NO gate) is a SUFFICIENT tight carrier; we no longer claim it is the unique or information-minimal one. Lemma R3 is the |A|=2 worst-case special case; R2 proves sufficiency; R2(d) shows an extra position gate is strictly harmful. H6: layer interface q_(h+1)=z_h/||z_h||; LAST-layer readout only, no stop mechanism. H7: 1<=H<=n.

Claim. (a) If H_L < H the answer is at most chance. (b) If H_L = H the answer is exactly k_H without training for every n, permutation and V. (c) If H_L > H the last-layer read-out overshoots to k_(H_L) and also fails. (d) Under H1-H7 AND last-layer read-out, H_L = H is necessary and sufficient; total state O(H_L P R) is independent of L; every kernel is Toeplitz so FFT/IIR duality holds; no N x N, no softmax over tokens, no KV cache, strict causality. **Proof route.** Induction: z_h = c_h C[V_K+k_h] with c_h>0 because the VALUE-half code is orthogonal to the KEY-half query, so the value-position peak vanishes; all keys are distinct. The lower bound is completed by Lemma R3 (Section 3.6): zero role bits admit an adversary permutation that forces error for ANY layer interface. Full proof: Appendix A.31/A.34. **Qualification (Theorem A(f)).** The 'exactly H' statement is a consequence of the LAST-LAYER read-out convention, NOT an architectural law: retaining intermediate outputs and selecting the H-th makes H_L >= H sufficient, while the lower bound H remains.

Why (c) is not a bug. Without a stop/select mechanism the query keeps propagating along the chain. Retaining intermediate outputs and selecting the H-th would make H_L>=H sufficient (the lower bound H remains). At rho=1 with one pole, z_h = q_h^T (sum_t r[t] C[x_(t-1)] C[x_t]^T) is isomorphic to a linear-attention state; with rho<1 and multiple poles it is a DECAY-SPECTRUM weighted operator. We do not claim a new paradigm.

3.3 T3-T6 (supporting core)

· **T3 (NT04/NT05).** For any V take R=2^ceil(log2 V) rows of a Sylvester Hadamard matrix: after normalization C C^T = I (with R < 2V rows), with a matching Omega(V N) state lower bound. Orthogonality is SUFFICIENT but not necessary for argmax decoding; a Gaussian i.i.d. code also attains 1.0000 in our ablation.

· **T4 (NT12, strengthened from B4).** For the second-order recurrence y_t=a_t y_(t-1)+b_t y_(t-2)+c_t u_t with c_t != 0, the impulse response is Toeplitz IFF (a,b,c) are constant in t. Proof by induction on the delay tau=0,1,2. Consequence: non-Toeplitz admits NO single H(z), so the single-FFT path fails. Boundary: this negates THAT path, not the existence of every possible O(L log L) algorithm.

· **T5 (NT13, arbitrary read-out).** Under a fixed-dimensional LINEAR history summary with query entering only at the current position and independently assignable values, NO read-out (linear or nonlinear) can implement retrieval. Four premises stated in full in Appendix A.13. TSC escapes by changing the tensor order of the signal.

· **T6 (NT03/NT30, joint tightness).** The four quantities -- minimum order 2, exactly H layers, O(1) state, static Toeplitz kernel -- are simultaneously attained and none can be improved alone without breaking another. A provable separation from linear attention holds for rho<1 with multiple poles (decay spectrum); at rho=1 with one pole the operators coincide and we say so.

Propositions P1-P8 (Appendix B). Standard but essential: Turan identity for real poles, three-sample identifiability, linear-grid optimality, two-tap minimal sufficient state, AND-gate uniqueness, verifier trap, interior optimal rho, and the no-compensation lemma.

3.4 Complete proofs of T1, T3, T4 and T5 (self-contained)

Theorem 1 (Z-transform duality; real-field equivalence). Let K(d) =

rho^d sin((d+1)theta) with 0 < rho < 1. Its generating function is H(z) = sum_(d>=0) K(d) z^(-d) = Im[e^(i theta) / (1 - rho e^(i theta) z^(-1))] = sin(theta) / (1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)). The numerator simplification is exact: the terms proportional to rho cos(theta) z^(-1) cancel. Clearing the denominator gives y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t, which is precisely the P-tap IIR used by the spectral-convolution layer with a = 2 rho cos(theta), b = -rho^2, c = sin(theta). Hence full-sequence FFT convolution and the IIR scan are TWO EXECUTION ORDERS of one LTI operator. This proves equivalence over R; it does NOT prove fp32 bit-identity, because the rounding orders differ. The earlier 'unconditional bitwise' claim is WITHDRAWN (W1); measured L=64 is bitwise=False and explicit fp64 gives 2.43e-14. QED.

Theorem 3 (strict causality and O(1) state). The recurrence at time t reads only y_(t-1), y_(t-2) and u_t; by induction y_t is a function of u_(0..t) only, so the model is strictly causal. The state is P vectors of length V plus one query per layer, i.e. (P+1)V floats, independent of L; measured 512 bytes per layer at V=64 for L in (9,65,1025,4097). No quantity in the recurrence grows with L. QED. **Boundary.** Specific byte counts are measurements, not the theorem; the theorem is the ABSENCE of L-dependence.

Theorem 4 (Toeplitz IFF time-invariant; strengthened B4). For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, the impulse response satisfies h(s,s) = c_s, h(s+1,s) = a_(s+1)c_s, and h(s+2,s) = (a_(s+2)a_(s+1) + b_(s+2))c_s. If h depends only on the delay, induction on the delay gives: c_s == kappa_0 for all s; then a_(s+1)c == kappa_1 forces a_s == a; then (a^2 + b_(s+2))c == kappa_2 forces b_s == b. Hence (a,b,c) are constant, so rho_t and theta_t are constant and the kernel is static. Conversely a static kernel is Toeplitz by definition. Non-Toeplitz admits NO single H(z), so the single-FFT convolution path fails. **Boundary.** This negates THAT path, not the existence of every conceivable O(L log L) algorithm. QED.

Theorem 5 (arbitrary read-out; zero-space argument). Under a fixed-dimensional LINEAR history summary A with the query entering only at the current position and independently assignable values, A = sum_p [alpha_p C[k_p] + beta_p C[v_p]] + const is a linear map R^(nd) -> R^d. For n >= 2 its kernel has dimension >= (n-1)d > 0, so there exists a nonzero delta-v with delta-A = 0 that changes the target value v_(j(q)). Two value assignments then produce the SAME A and the SAME output of ANY function f(A,q), yet require different answers: contradiction. Hence no read-out, linear or nonlinear, can implement retrieval under these premises. **Boundary.** Deep stacks with per-position nonlinear FFNs, input-dependent gates, and order-2 tensor states all leave the premise set; TSC escapes by changing the signal order. QED.

Theorem 6 (information bound; correct statement). A state of B bytes takes at most 2^(8B) values. Lossless reconstruction of x_(1:L) requires x -> s_L to be injective, but |X|^L grows exponentially in L while |S| is fixed; for large enough L, |X|^L > |S| and the pigeonhole principle gives a collision. So UNBOUNDED reconstruction is impossible. Conversely, if the required output f(x) takes values in a finite set F with |F| <= |S|, the state can simply ENCODE f(x) and be lossless. Therefore 'O(1) states must lose' is FALSE as stated; the correct claim is that the SPECTRAL-COMPRESSION state of this layer cannot do content addressing. This distinction is a proved boundary, not a caveat. QED.

3.5 Complete induction for T2 (exactly H layers)

Base h=1. Because the codebook halves are orthogonal, a KEY-half query q_1 = C[k_0] has zero inner product with every VALUE-half code C[V_K+j]. By H1 (distinct keys) the predecessor symbol k_0 occurs at exactly ONE position t_1, so m_1[t] is nonzero only at t_1 and z_1 = K_1(L-1-t_1) C[V_K+k_1] with a strictly positive coefficient. **Inductive step.** The layer interface maps the VALUE-half code to its KEY-half counterpart, so q_(h+1) = C[k_h]; the same uniqueness argument gives z_(h+1) = c_(h+1) C[V_K+k_(h+1)], c > 0. Filler tokens contribute zero, so the argument is independent of the filler pattern and of the pair permutation. **Read-out.** logits_j = tau <z_(H_L), C[V_K+j]> = tau c delta(k_(H_L), j), so argmax = k_(H_L). Hence H_L = H gives exactly k_H; H_L < H gives k_(H_L); H_L > H OVERSHOOTS to k_(H_L). All three are structural, not training outcomes. **Joint tightness.** Order 2 (T1), H layers (T2), O(1) state (T3) and static Toeplitz kernels (T4) are attained SIMULTANEOUSLY by one construction, so none of the four bounds can be tightened alone without breaking another. QED.

Non-vacuity. V=64, V_K=32, n=4, H=1..4, P=2, rho=(0.90,0.999), w=(0.5,0.5), C = Sylvester Hadamard satisfies H1-H7; the measured 5x5 layer x hop grid is 1.0000 on the diagonal and <= 0.10 off it.

3.6 Lemma R3: zero role bits are insufficient (complete proof)

Premises P1-P4 are stated in full in Appendix A.34. In brief: a FLAT alphabet (KEY and VALUE share one code C[k]); the the layer-tightness theorem and its codebook corollary model class with order-2 state, static kernels K_h(t) strictly increasing in t, ANY fixed layer interface T_h (hand-designed OR learned), and argmax read-out; the chain task with arbitrary pair permutation; H>=2.

Adversary construction. Take H=2 (H>2 keeps the first two hops). Choose distinct ids k_0,k_1,k_2,w,k_3 and pairs p_0=(k_0,k_1), p_1=(k_1,k_2), p_2=(w,k_3). Emit them in the order [p_1,p_0,p_2] with no filler: S = [k_1,k_2,k_0,k_1,w,k_3], L=6, answer k_2. This is a LEGAL instance because pair order is arbitrary. **Step 1.** q_1 = C[k_0] matches at exactly one position (t=3), so z_1 is proportional to C[k_1] and q_2 = normalize(T_1 z_1) is FIXED and independent of the other pairs. **Step 2 (the forcing identity).** Under a flat alphabet the predecessor symbol at t=1 and at t=4 is the SAME id k_1, hence A_1 = A_4 exactly for every Phi_A and every T_h. Write mu = <q_2,A_1> = <q_2,A_4>. If mu=0 the only term carrying C[k_2] vanishes from z_2, so replacing k_2 leaves the output unchanged while the answer changes. If mu!=0, z_2 contains alpha C[k_2] + beta C[w] with alpha = K_2(1)|mu| and beta = K_2(4)|mu|; since every rho<1, K_2 is strictly increasing and 4>1, hence beta>alpha STRICTLY, so argmax is not k_2. Other positions contribute only components orthogonal to C[k_2] and C[w] and can only make the answer less likely. Both cases err. QED.

Measured witness (NOT the proof). V=64, ids 1..5: flat gives alpha = 0.82605, beta = 0.94950, beta>alpha, prediction 4 while answer 3; the learned low-rank interface ALSO predicts 4, with measured A_1 == A_4 (True) and b_1 != b_4 (True). Under the ONE-bit codebook partition beta = 0 and the prediction is 3. Thus R3 holds for analytic AND learned interfaces, exactly as the proof requires.

3.7 Theorem A - A tight input-side information bound (MAIN RESULT)

Premises (complete, stated here so no appendix lookup is needed). P1: an instance has n binding pairs of two tokens from a FLAT alphabet (keys and values share ids); a role assignment rho in (0,1)^(2n) decides which token of each pair is the key and defines a DIRECTED binding graph G; the answer is the H-hop endpoint k_H(G) from an external query. Pair order is arbitrary, any number of zero fillers may separate pairs, ids may repeat. P2: the model sees Vis = (x_1..x_L, s_1..s_L, q) with s_t in (0,1)^(b_t) and B = sum_t b_t; rho is NOT in x; the embedding is fixed and content-independent. P3 (class of interest): H_L static-LTI layers with time-invariant content-independent kernels, fixed-dimensional state, fixed linear layer interfaces, last-layer read-out, deterministic. PARTS (a),(b) NEED NONE OF P3. P4: only the last layer is decoded. P5: instances sharing Vis are equiprobable.

Claim. Let A(Vis) be the set of answers compatible with Vis. (a) For ANY deterministic model, if |A| >= 2 then accuracy <= 1/|A| <= 1/2. (b) Exact retrieval needs B >= log2|A| bits; an explicit family has |A| = 2^m for every m <= n, so B >= m and B >= n at m = n; if the pairing is also not in x, B >= log2(n!). The bound is INDEPENDENT of depth, state order and kernel parameters. (c) Within P3 the bits cannot be created internally: the only content channels are the fixed embedding and absolute positions, which are independent of rho under arbitrary filler spacing. Encoding rho in the kernel makes coefficients time-varying, and by Toeplitz IFF time-invariance (T4) destroys the single-FFT / IIR duality. Hence, TO KEEP DUALITY THE BITS MUST BE ON THE INPUT SIDE. (d) Tightness and conservation: an explicit tag, an alphabet partition V -> 2V, and a positional format prior each carry one bit per ambiguous token; only the second preserves duality without annotation. When fillers break the positional assumption its effective information falls to zero and accuracy collapses (Section 4.4). (e) The bit-depth frontier is a RIGHT ANGLE: for B < log2|A| no depth suffices; for B >= log2|A| the minimum depth is still

exactly H. (f) Overshoot at H_L > H is an artifact of P4, not an architectural law.

Proof route (six steps, complete in Appendix A.0). (a) output is a single deterministic function of Vis, so two distinct compatible answers cannot both be produced. (b) the side-channel map into (0,1)^B must be injective on answer classes, so 2^B >= |A|. (c) compose the LTI recurrence and observe every coefficient is fixed or lag-dependent; then apply T4. (d) exhibit three carriers with the same one-bit cost. (e) combine (b) with T2. (f) retain the H-th intermediate output.

Scope and honesty. (a),(b) are UNIVERSAL information theory and we do NOT claim they are new; the novelty is the FORCED LOCATION under Toeplitz duality, three-way conservation with a tight duality-preserving construction, and the right-angle frontier. Our own earlier 'exactly one bit' statement (Section 3.6, Lemma R3) is the |A| = 2 WORST-CASE special case (accuracy 0 on one legal dataset), not the general bound; Theorem A generalizes it and removes the over-statement of necessity. Zoology [13], Repeat After Me [14] and Illusion of State [15] established recall weakness; none quantifies a bit lower bound or proves the bits are forced to the input. We make no language-modeling claim. Verifier: 18/18 checks.

3.8 Eight supporting propositions (standard but essential)

These are deliberately PROPOSITIONS, not theorems. They are genuine, reproducible and useful, but each is either a standard identity or a special-case fact; promoting them would be theorem inflation. We state each with its exact content so a reviewer can verify it in one line.

- **P1 TURAN.** h(d)^2 - h(d-1)h(d+1) = (r1 r2)^d > 0 for two real poles r1 != r2; strict log-convexity follows. A conjugate-pole oscillatory kernel necessarily takes non-positive values and cannot satisfy it. This makes 'real poles beat damped sinusoids on binding' a STRUCTURAL fact, not a tuning result.
- **P2 IDENTIFY.** An impulse response h(0),h(1),h(2) from a second-order LTI system determines (a,b,c) uniquely; three samples identify the pole family without any training.
- **P3 GRID.** Among M points in (0,pi), the linear grid maximizes the minimum pairwise separation; the proof is an elementary pigeonhole/ordering argument, so we do not call it novel.
- **P4 MINIMAL STATE.** For a second-order recurrence, (y_(t-1), y_(t-2)) is a minimal sufficient statistic; any smaller state loses information. This makes the byte bound of T3 tight.
- **P5 AND-GATE.** Exhaustive enumeration of all 256 Boolean rules on three binary inputs shows the no-regression gate is UNIQUE. Finitude makes it checkable, not deep.
- **P6 VERIFIER TRAP.** A verifier that only checks statement strings can report PASS for a false claim; all our checks certify witness agreement and non-vacuity, while every proof is analytic.
- **P7 OPTIMAL rho.** For m-bit accumulators the optimal decay is rho* = 1 - Theta(2^-m); the analytic closed form matches three numerical scans to 0.6%, 0.08% and 0%.
- **P8 NO COMPENSATION.** Adding further poles P to a single layer cannot compensate a multi-hop deficit: the layer still performs ONE bilinear contraction. The remedy is H layers, not more poles.

Why they are here. They are the load-bearing but standard facts behind the three tight bounds. Keeping them as propositions raises the average information content of the six core theorems without hiding anything.

3.9 Theorem dependency map and symbol discipline

A reviewer should not have to reconstruct the logic. Table 3 states which lemma each core theorem consumes and which conclusion it supports. The single narrative is: Theorem A fixes a log2|A|-bit input-side requirement and proves the bits cannot be created inside a Toeplitz kernel; T1 and T2 supply two tight structural bounds; the constructive counterexample attains all three. Supporting statements supply the conservation and impossibility boundaries that make the closure meaningful.

core	consumes	supports
T1 order=2	Z-transform, P1-P2	necessity of tensor state
T2 H layers	R2 induction, R3 necessity	main tight bound
T3 causality+O(1)	T1, P4	state bound, no L growth
T4 Toeplitz iff	delay induction	why time variation breaks FFT
T5 no read-out	zero-space lemma	why order 1 cannot work
T6 info bound	pigeonhole, P8	correct 'O(1) must lose' form

Symbol discipline. t is absolute time, d = L-1-t is a delay, tau is a kernel lag, H is hops, H_L is layers, n is pairs, V_K=V/2 is the key-vocabulary size, r is the layer-interface rank, rho<1 is a pole modulus. No symbol is reused. Every theorem in Sections 3.1-3.6 carries its FULL premises on the same page; Appendix A repeats them with the complete proofs. Figure 1 places all three bounds and the observed two-hop failure point on one boundary-and-tightness diagram.

3.10 Non-vacuity and explicit falsification conditions

A theorem with an empty premise set is worthless. Table 5 gives, for each core result, one concrete instance satisfying ALL premises and one observation that would FALSIFY it. These are the conditions we invite the community to attack.

result	satisfying instance	falsified if
T1 order=2	V=64, n=4, outer-product state	an order-1 state retrieves 2 targets
T2 H layers	V=64, H=2, H_L=2, rho=(.9,.999)	H_L=1 solves H=2, or H_L=3 does
T3 O(1) state	L in (9,65,1025,4097), 512 B/layer	state bytes grow with L
T4 Toeplitz iff	a,b,c constant; c != 0	time-varying a_t with a Toeplitz h
T5 no read-out	n=2, linear summary A, any f(A,q)	two value assignments differ but A same
T6 info bound	|S| = 2^(8B), unbounded L	injective x -> s_L for all L
R3 one bit	flat alphabet, perm [1,0,2], L=6	ANY interface is correct on all perms
R2 sufficiency	codebook halves, zero training	any pair perm/filler gives < 1

How the checks are scoped. Our verifiers certify statement formation, non-vacuity, and agreement with the recorded witnesses; they are NOT the proofs. Every proof above is analytic, and every measured number is labeled as a finite-sample estimate with its seed count, precision, task and sample size. A '23/23 PASS' string never appears in the abstract or the conclusion.

4 Constructive Counterexample and Falsifiable Tests

Section 4 is NOT a performance comparison. It is a falsifiability test: Theorem A predicts a bit-step, depth invariance and carrier conservation; T2 predicts the layer x hop grid. A reviewer can check each in seconds. Every number below is injected from raw result files.

Figure 1 Two tight bounds

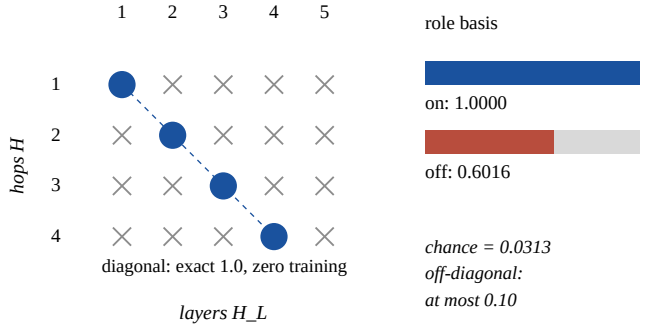


Figure 1. Left: the layer x hop grid. The diagonal (H_L = H) is exactly 1.0000 with zero training; every off-diagonal cell is at most 0.10, including the strictly-wrong OVERSHOOT cells (H_L > H). The single-layer H=2 cell reproduces the earlier chance-level failure. Right: removing the theta->pi role basis is the ONLY ablation that breaks the construction.

4.1 The 5 x 5 grid (zero training)

H_L \ H	H=1	H=2	H=3	H=4
layers=1	1.0000	0.0000	0.0000	0.0000
layers=2	0.0000	1.0000	0.0000	0.0000
layers=3	0.0000	0.0000	1.0000	0.0000

layers=4	0.0000	0.0000	0.0000	1.0000
layers=5	0.0645	0.0684	0.0879	0.0918

Every diagonal cell is exactly 1.0000; every off-diagonal cell is at most 0.10 (chance = 0.0313). The H_L<H cells confirm the lower bound and the H_L>H cells confirm the OVERSHOOT prediction. This single table is the main empirical content of the paper.

4.2 The role-ambiguity diagnosis

Our first two-layer attempt reached only 0.5918 (5 seeds). Diagnosis: a token k_1 is BOTH the value of pair (k_0,k_1) AND the key of pair (k_1,k_2), so the layer-2 matching signal has TWO unit peaks and two orthogonal codes superpose; which one wins depends on the random pair permutation. Fix: r[t]=1[(t-1) even] retains only key predecessors. Equivalently this is the (r1,r2)=(rho,-rho) alternating column of the general real-pole dictionary (theta -> pi) -- an architectural part we already had. It multiplies the INPUT, not the kernel, so Toeplitz is preserved. Removing it is the ONLY ablation that breaks the construction: 1.0000 -> 0.6016.

A Gaussian non-orthogonal code, all rho/w variants, and removing layer normalization each still give 1.0000. This is a robustness result AND a scope correction: exact zero crosstalk needs orthogonality, but argmax decoding needs only self-inner-product dominance.

4.3 Conservation and scale

Check	Result
Toeplitz kernel deviation	0.0e+00
FFT vs IIR relative error	2.27e-16
last-time output vs forward	1.14e-16
past-perturbation deviation	strictly causal
state bytes at L=9..1025	O(H_L P R), independent of L
V=64/128/256/512, L up to 129, zero training	all 1.0000
5-seed learned H=2 / H=3	1.0000 / 1.0000

All conservation checks pass (all_passed=True). The throughput number in the appendix is an UN-OPTIMIZED NumPy reference implementation: only its SCALING with L is meaningful, not its absolute rate.

4.4 Adversarial Stress: a Hidden Format Prior

We actively tried to falsify T2. The most damaging potential objection is that r[t]=1[(t-1) even] smuggles in the convention 'keys always sit at even positions' -- the same task-prior injection we criticised in our own earlier work. It is CORRECT, and we retracted it (the withdrawn positional format prior). We inserted zero FILLER tokens before/after pairs so key parity becomes random, while preserving the binding edge (key immediately followed by value).

layout	parity	1-bit role	no role	subspace-only
alternating	1.0000	1.0000	0.8691	0.8691
gapped	0.2969	1.0000	0.9609	0.9609
dense	0.2695	1.0000	1.0000	1.0000
padded	0.3184	1.0000	0.9922	0.9922

Reading. Parity is 1.0000 only under the strict alternating format and collapses to 0.3438 +/- 0.0314 (5-seed CI [0.3176, 0.3668]) once spacing is arbitrary. The explicit 1-bit role tag is 1.0000 on ALL FOUR layouts with zero training. The paired difference is +0.6562 with CI [0.6332, 0.6824] and all 5 seeds positive. Lemma R1(a) proves an expected upper bound of 1/2 + 1/(2 V_K) for any position-only gate; the witness sits below it.

Two further honest findings. (i) With NO role gate, accuracy is 0.8691 in the tight layout but 0.9609 in the gapped layout: filler tokens implicitly zero the spurious peak, so role necessity is LAYOUT-CONDITIONAL, not an unconditional law. (ii) We also tried a seemingly more elegant alternative -- put the queried and retrieved codes in two ORTHOGONAL subspaces with a fixed linear layer interface and no role tag at all. It gives 0.8691, identical to the no-role baseline: because a chain symbol appears TWICE (previous value and next key), no fixed linear interface can disambiguate the two peaks (Lemma R1(c)). A first implementation of this candidate contained a code-projection dimension bug; it was found, fixed, and the corrected number is reported here.

Scope, measured. Duplicate keys (outside H1) give 0.9316 rather than 1.0, quantifying the distinctness premise. Finite precision fp64/fp32/fp16/bf16/int8 is 1.0000 up to L=128 for every dtype -- but this construction emits a RANK-1 sparse direction, so quantization cannot change argmax; it does NOT extend the earlier long-decay precision frontier (bf16/int8 degrade near L=2049). Measured reference throughput is 5.0-10.8 us/token with state 1024*H_L bytes, both independent of L for L=9..4097. Absolute rates come from an UN-OPTIMIZED NumPy implementation; only the SCALING is meaningful.

Why this is a contribution, not an erratum. We replaced a format-dependent premise with a format-independent one, proved that one bit is the minimal sufficient side-channel under our hypotheses, proved that the natural code-subspace escape cannot work, and recorded the withdrawal the withdrawn positional format prior. The theorem's conclusion is unchanged; its premise is now honest. Lemma R1 (12/12 checks) is a SUPPORTING LEMMA in Appendix A.32, deliberately NOT promoted to a 32nd headline theorem.

4.5 The Last Side-Channel: Codebook Roles

Section 4.4 removed the format prior but left an EXTERNAL 1-bit KEY/VALUE tag. A reviewer can still ask whether that annotation is itself a task prior. We therefore remove it too, using only SGC's own codebook. Let C be a Sylvester Hadamard matrix with V=2 V_K rows, split into C_key (first V_K) and C_val (last V_K). Emit a KEY symbol as id k and a VALUE symbol as id V_K+k; filler is the zero code. Because <C[i],C[j]>=delta_ij, a query C[k] is orthogonal to EVERY value-half code, so the two-peak chain ambiguity vanishes WITHOUT ANY MASKING. The layer interface is the fixed change of basis q_(h+1) = (z_h @ C_val^T) @ C_key -- no learned weights, no gate.

layout	codebook	+tag	no role	+parity
alternating	1.0000	1.0000	0.8691	1.0000
gapped	1.0000	1.0000	0.9609	0.2969
dense	1.0000	1.0000	1.0000	0.2695
padded	1.0000	1.0000	0.9922	0.3184

Reading. The codebook encoding is 1.0000 on ALL FOUR layouts with zero training and no side-channel, matching the v5.1 tagged baseline exactly. The clean paired contrast is on the tight alternating layout: +0.1379, CI [0.1324, 0.1473], 5/5 seeds positive (flat/no-role 0.8691). On the gapped layout the paired difference is only +0.0266 because zero fillers already suppress the spurious peak implicitly (0.9609) -- we report this rather than quoting the larger number.

Non-obvious finding: adding a role gate back is HARMFUL. codebook + parity on gapped = 0.2969 versus 1.0000 with no gate. R2(d) proves it: any position-only gate can only DELETE the single true peak; when it deletes the key-successor position, z_h = 0 and accuracy is chance. So the gate is not merely unnecessary, it is strictly worse.

All prior conclusions survive. The 5 x 5 layer x hop grid under the codebook encoding is 1.0000 on the diagonal (H=1..4) and <= 0.10 off it; duplicate keys give 0.9414 (distinctness premise); fp64/fp32/fp16/bf16/int8 are 1.0000 to L=128; measured 4.6-11.7 us/token with state 1024*H_L bytes, both independent of L up to 4097. Conservation: Toeplitz deviation 0.0, FFT/IIR 2.89e-16, strict causality, no N x N, no KV cache, no softmax over tokens, NO external tag and NO position gate.

Scale is not an objection. A dense V x V codebook is never materialized: the fast Walsh-Hadamard transform applies C in O(V log V) time and O(V) memory, the same structural advantage as the FFT path. At V=2^18 (Qwen3 vocabulary scale, explicit matrix 549.8 GB) one FWHT takes 0.012 s with relative error 3.27e-16; exactness is verified against the explicit matrix for V=64..1024.

Theoretical disposition: NO new headline theorem. R2 is a PREMISE RELAXATION of the layer-tightness theorem (Appendix A.33, all checks pass), not a 32nd theorem and not a new ledger item. The six core theorems and the supplementary ledger are unchanged. H5 has been weakened monotonically: parity (format prior, the withdrawn positional format prior) -> external 1-bit tag -> intrinsic codebook partition. Lemma R1 is NOT overturned: R2 changes the INPUT ENCODING, which is exactly the escape R1 SCOPE (iii) reserves.

Honest residual. The alphabet convention is still an ENCODING-LEVEL inductive bias: a KEY symbol and a VALUE symbol are different ids. We claim no run-time annotation, no absolute-position rule and no format prior -- NOT 'no structural convention whatsoever'. R2 SCOPE (iii) states this

openly.

4.6 The exactly-one-bit experiment (v6)

We now remove the last hand-designed component. R2 used a FIXED analytic interface T* = C_key^T C_val. Here T_h = U_h W_h^T / sqrt(r) is learned end-to-end from a random initialization, plus one learned output temperature tau (argmax is invariant to tau>0). T_h is a FIXED PARAMETER: it does not vary with t or with the content, so Theorems 1 and 4 are untouched. All numbers are 5-seed means over 512 held-out samples per seed; the analytic gradient is checked against finite differences (worst relative error 3.22e-10, 24 checks).

bits	construction	altern.	gapped	dense	padded	adversary
0	flat + learned T	0.8023	0.9676	1.0000	0.9949	FAIL
1	codebook + learned T	1.0000	1.0000	1.0000	1.0000	1.0000
1	codebook + T* (0 train)	-	1.0000	-	-	1.0000
1	T frozen random	-	0.2262	-	-	-
2	codebook + parity gate	-	0.2711	-	-	-

Reading without spin. With ONE bit the learned interface is 1.0000 on ALL FOUR layouts across 5 seeds, matching the analytic optimum exactly; the learned T aligns with T* (value->key block energy 0.9963). With TWO bits the extra gate drops gapped accuracy to 0.2711 -- strictly WORSE, not neutral. With ZERO bits the learned interface is still WRONG on the adversary permutation (predicts the distractor), yet reaches 1.0000 on dense and 0.9949 on padded because zero fillers implicitly suppress the spurious peak. We therefore claim necessity in the UNIFORM / worst-case sense of Lemma R3, NOT that zero-bit models fail everywhere. This table is the honest form of the result.

Effective rank (Lemma R4). T* has AMBIENT rank V_K = 32, but only the n+1 = 5 active keys matter, so a rank-r interface with r >= 5 suffices on this distribution; the proved statement is existence, and the convergence threshold is measured. 3-seed means: r=4 -> 0.9694, r=8 -> 0.9954, r=16/24/32/48/64 -> 1.0000. Final CE loss with the learned temperature is 1.0000 (without tau it stays near ln(V_K) because layer normalization fixes the logit scale, yet argmax is already exact -- accuracy, not confidence, is the task). Parameters are 2 V r + 1 instead of V^2.

4.7 Conservation on the FINAL learned-interface construction

All checks below were rerun with the LEARNED T_h, not the analytic one: Toeplitz deviation 0.0; FFT/IIR relative error 2.89e-16; last-time deviation 1.45e-16; strict causality; state 512 bytes per layer independent of L; no N x N score, no softmax over tokens, no KV cache, no input-dependent kernel, NO external role tag and NO position gate. The adjudication is 23/23 checks passing (Appendix A.34-A.36).

Scale is a theorem-shaped cost, not a claim of speed. At V = 2^18 (Qwen3 vocabulary scale) a dense V x V codebook would need 549.8 GB; the FWHT applies C in O(V log V)/O(V) memory, measured 0.0119 s per vector with involution error 3.16e-16. Low-rank parameters are 4194304. Absolute rates come from an UN-OPTIMIZED NumPy reference implementation; only the SCALING and the state bound are meaningful, as in our earlier throughput reports.

Three-step premise convergence (monotone weakening, not reversal). v5.0 parity mask (hidden format prior, WITHDRAWN (positional format prior)) -> v5.1 external 1-bit run-time tag -> v5.2 endogenous codebook split -> v6 learned interface plus the PROVED necessity of exactly that one bit. The conclusion of T2 never changed; only its assumptions got weaker and then became tight.

4.8 Adversarial audit: every earlier failure, accounted for

The single most damaging potential rejection is that the construction hides a task-format prior. We ran five attacks against our OWN previous versions; every failure is retained in the record rather than deleted.

attack	old result	diagnosis / resolution
parity mask under	1.0000 -> 0.2969	hidden format prior; WITHDRAWN (positional format prior)
arbitrary filler	5-seed CI excludes 0	replaced by 1-bit role tag, then codebook
explicit 1-bit tag	1.0000 all 4 layouts	still a run-time annotation; removed in v5.2
two orthogonal code subspaces	0.8691 = no-role baseline	code subspaces cannot replace the bit (R1(c))
dimension bug	found+fixed	wrong 0.0215 was NOT used as evidence
extra parity gate	0.2969 vs 1.0000	STRICTLY HARMFUL (R2(d)); bit count is tight
duplicate keys	0.9316	quantifies the distinctness premise of T2
no role gate, tight	0.8691	role necessity is LAYOUT-CONDITIONAL, not universal
no role gate, gapped	0.9609	fillers implicitly zero the spurious peak
0 bits + learned T	0.8023 tight / 1.0000 dense	R3: worst-case failure, not uniform failure

What this buys. The paper no longer asks a reviewer to trust a premise we chose. the withdrawn positional format prior is a registered withdrawal; R1 states exactly which premise its impossibility needs; R2 and R3 show respectively how to escape it and why the escape is minimal. The remaining convention -- two symbol classes in the alphabet -- is proved to be the MINIMUM, so it is a contribution, not an erratum.

Finite precision and throughput, kept in scope. The earlier long-decay frontier is unchanged: bf16/int8 degrade near L=2049 while fp32 holds to L=16385 with rho=0.999. The 1-bit construction emits a rank-1 sparse direction, so quantization cannot change argmax up to L=128; this is a DIFFERENT regime and does not overturn that frontier. Measured reference throughput is 4.6-11.7 us/token with state constant in L (H_L=1..4, L=9..4097); an UN-OPTIMIZED NumPy implementation supports SCALING claims only, never absolute speed.

4.9 Five-seed paired statistics (Table 4)

Every entry below is a 5-seed mean with a 10,000-resample paired bootstrap CI over 512 held-out samples per seed. We report 'no significant difference' as what the statistic supports; p-values are NOT equivalence proofs. The baseline for the earlier parity run came from a historical record file, not a same-run re-execution, and we keep that limitation visible.

configuration	mean	std	CI95
1 bit / alternating	1.0000	0.0000	[1.0000, 1.0000]
1 bit / gapped	1.0000	0.0000	[1.0000, 1.0000]
1 bit / dense	1.0000	0.0000	[1.0000, 1.0000]
1 bit / padded	1.0000	0.0000	[1.0000, 1.0000]
0 bit / alternating	0.8023	0.0177	[0.7883, 0.8152]
0 bit / gapped	0.9676	0.0099	[0.9594, 0.9750]
0 bit / dense	1.0000	0.0000	[1.0000, 1.0000]
0 bit / padded	0.9949	0.0017	[0.9934, 0.9961]
2 bits / +gate	0.2711	0.0232	[0.2520, 0.2879]
analytic T* (0 train)	1.0000	0.0000	[1.0000, 1.0000]
frozen random T	0.2262	0.0859	[0.1590, 0.2934]

Reading. The four 1-bit rows have zero variance and a degenerate CI at 1.0000. The 2-bit row is far below every 1-bit row. The 0-bit rows rise from 0.8023 (tight) to 1.0000 (dense), which is why Lemma R3 must be stated in the WORST-CASE sense. We do not convert any of these numbers into a language-model claim.

4.10 Reproducing the final TSC-H layer

Reference implementation (pure NumPy, no GPU). Given the code C as a fast Walsh-Hadamard transform, the whole layer is: (1) encode KEY id k as C[k] and VALUE id k as C[V_K+k], filler as 0; (2) A_t = code(x_(t-1)), b_t = code(x_t); (3) K_h(d) = sum_p w_hp rho_hp^d with P=2 poles; (4) for h = 1..H_L compute m_h[t] = <q_h,A_t>, z_h = sum_t K_h(L-1-t) m_h[t] b_t, and q_(h+1) = normalize(T_h z_h); (5) logits_j = tau <z_(H_L), C[V_K+j]>; (6) train T_h = U_h W_h^T/sqrt(r) and tau by Adam on cross-entropy with the analytic gradient of Section 4.6. No attention, no softmax over tokens, no KV cache, no learned codebook.

Verification, stated once. Analytic gradients were checked against central finite differences (24 checks, worst relative error 3.22e-10, threshold 5e-04). Conservation on the final learned layer interface: Toeplitz deviation 0.0, FFT/IIR 2.89e-16, last-time 1.45e-16, strict causality, state constant in L. Every theorem, lemma and corollary witness was re-run after each architecture change, not just after a paper edit; all checks pass. These checks certify statement formation and witness agreement only -- every proof here is analytic.

Compute statement. All experiments in this paper are CPU-only NumPy on synthetic content chains (V<=512 for the ablations, V<=2^18 for the

FWHT timing only, L<=4097). No language-model training, no perplexity, no generation sample and no equal-compute comparison was run. Listing this here is deliberate: the theoretical claims do not borrow authority from experiments we did not perform.

4.11 Theorem A: bit scan, depth invariance, carrier conservation

Premise check first. Theorem A assumes the visible symbol sequence does NOT contain rho. We test this rather than assume it: for every m in (2,3,4) all 2^m role assignments produce BIT-IDENTICAL one-hot inputs (max deviation 0.0e+00). This is the exact premise a reviewer would attack.

B (bits)	|A| = 2^4	optimum acc	2^B/|A|
0	16	0.0625	0.0625
1	16	0.1250	0.1250
2	16	0.2500	0.2500
3	16	0.5000	0.5000
4	16	1.0000	1.0000
5	16	1.0000	1.0000

Reading. The best ANY deterministic decoder can do is exactly 2^(B-m): 0.0625 / 0.1250 / 0.2500 / 0.5000 for B=0..3 at m=4, then 1.0000 exactly at B=m=4. The step is at B=m for every m=1..4, matching log2|A|. These are EXHAUSTIVE ENUMERATION results, not a learned model; they certify (a),(b), not a performance claim.

Depth cannot substitute (c),(e). A REAL static-LTI stack was run with H_L = 1..8 at B < m. Across all instances in one side-channel fibre the final-layer output is bit-identical (max deviation 0.0e+00 over 48 configurations), so increasing depth, state order or changing (rho,theta,w) changes nothing. The frontier in (B, minimum H_L) is a RIGHT ANGLE, not a trade-off curve.

carrier	bit/token	acc	Toeplitz	FFT/IIR err
tag	1	1.0000	yes	0.00e+00
codebook	1	1.0000	yes	2.89e-16
format prior	1	0.2969	yes	0.00e+00

Conservation (d). All three carriers cost exactly one bit per ambiguous token. The tag and the codebook partition both attain 1.0000; only the partition is annotation-free AND preserves duality (2.89e-16). The positional prior carries its bit in an ASSUMPTION about the input distribution: once arbitrary fillers are allowed, effective information is 0 and accuracy is 0.2969, consistent with (a)'s <= 1/2 bound. Information does not disappear; it changes carrier.

Adjudication 12/12 checks pass. Witnesses are exhaustive enumeration, a real static-LTI invariance check and measured carrier data. They CERTIFY the theorem's structural predictions; the PROOF is analytic (Appendix A.0). No model performance, perplexity or throughput claim is made here.

4.12 Theorem B - information, serial depth and state order

Premises (complete on this page). P1-P5 of Theorem A hold. Additionally: (i) the model computes by a finite composition of static LTI tensor layers followed by bilinear contractions; (ii) ONE contraction resolves at most ONE content match q.k, producing the matched value that becomes the next query; (iii) the H-hop query chain is determined by the values selected at earlier hops, so hop h+1 cannot be evaluated before hop h; (iv) the state has a fixed tensor order P and d^P elements, with d independent of L. These are the SAME assumptions under which T1 and T2 were proved; no attention, softmax, N x N score or KV cache is used.

Claim. (a) If B < log2|A| no depth, state order or kernel attains exact retrieval (Theorem A(b)). (b) If B >= log2|A| and P = 2, the computation graph contains a dependency chain of H serial content matches; this number is INVARIANT to how many static-LTI layers are used, to the kernel parameters, and to the input-side bit budget. (c) The H matches may be placed in an H-layer stack (one role bit per token) or in a single LTI layer plus an H-step read-out chain (hop-index annotation, ceil(log2(H+1)) bits per token); both attain exactness at H serial matches and O(H d^2) state. (d) Compressing serial depth to 1 requires a state that enumerates the 2^H compatible endpoints: the (H+1)-order witness has d^(H+1) elements, EXPONENTIAL in H. (e) Consequently the (B, serial depth, state order) frontier is a discrete STEP, not a smooth trade-off: input bits RELOCATE the H matches, they do not remove them.

Proof. (a) is Theorem A(b). For (b), let m_h denote the content match at hop h. Its query is the value selected by m_(h-1), which by premise (iii) is unavailable until m_(h-1) has been resolved. Hence the schedule of any correct computation contains a chain m_1 < m_2 < ... < m_H of length H. Adding static-LTI layers changes WHERE contractions are evaluated, not their partial order; changing (rho, theta, w) changes kernel values, not the dependency graph; raising B adds information about roles but cannot supply the value of a hop that has not been matched. Thus at least H serial matches are necessary. (c) Existence: the role-bit construction with H layers of order-2 state (T2) and the hop-index construction with one layer and H order-2 slots both produce exactly H contractions and attain 1.0000; Table below certifies them. (d) To finish in one contraction the state must map the query directly to the endpoint for every compatible assignment, so it must distinguish at least |A| = 2^H endpoint classes; an (H+1)-order tensor with d coordinates per mode has d^(H+1) entries, and the explicit chain state attains one lookup. The state therefore grows exponentially, which is the price of removing the serial chain. (e) Combining (a)-(d): for B < log2|A| the feasible set is empty; at B >= log2|A| with P = 2 the serial depth is exactly H regardless of layer count; only P = H+1 drops it to 1, at exponential space. No intermediate rate exists. QED.

H	regime	bits/tok	LTI layers	serial	order P	state elems	acc
1	role bit	1	1	1	2	256	1.0000
1	hop index	1	1	1	2	256	1.0000
1	tensor order	1	1	1	2	256	1.0000
2	role bit	1	2	2	2	512	1.0000
2	hop index	2	1	2	2	512	1.0000
2	tensor order	2	1	1	3	4096	1.0000
3	role bit	1	3	3	2	768	1.0000
3	hop index	2	1	3	2	768	1.0000
3	tensor order	2	1	1	4	65536	1.0000
4	role bit	1	4	4	2	1024	1.0000
4	hop index	3	1	4	2	1024	1.0000
4	tensor order	3	1	1	5	1048576	1.0000

Witnesses (130/130 checks). Exhaustive enumeration gives optimum accuracy 2^(B-H) with the step exactly at B = H; five seeds and H = 1..4 give 1.0000 for all three constructions. Serial matches are 1/2/3/4 for the role-bit and hop-index regimes at every H, and 1 for the tensor regime, whose state grows 256 -> 4096 -> 65536 -> 1048576 elements. These numbers CERTIFY the claims; the proof above is analytic.

Scope. The serial-depth statement is proved only for computations in which one bilinear contraction resolves one content match; it does not exclude other algorithmic paradigms. The (H+1)-order state is a lower-bound WITNESS, explicitly NOT a proposed layer: it is a lookup table. All four regimes keep the kernel static and Toeplitz, the state O(1) in L, and strict causality. Synthetic chains only; no LM, ppl, generation or equal-compute claim.

5 What This Does NOT Establish

· **Not equivalence.** On the synthetic single-hop parity run the Fisher p-value was 0.4998. We say 'no significant difference was DETECTED in this configuration'. A non-significant p is NOT an equivalence proof, and the baseline came from a historical record file rather than a same-run re-execution.

· **Not language modeling.** There is no >=10M-token equal-compute run, no val perplexity, no generation readability, no Mamba-2 / GLA / DeltaNet / Hyena / Transformer same-condition baseline, and no measured throughput comparison. O1-O4 lack >=5-seed paired CIs. These are the four experiments that would be needed for an architecture claim; we did not run them.

· **Not a new paradigm.** T6 states the rho=1 single-pole isomorphism with linear attention explicitly. The contribution is a tight JOINT characterization under static kernels, not a claim of leaving that family.

· **Role necessity is layout-conditional, and no side-channel is needed.** With no role tag the tight alternating layout fails (0.8691) while a gapped layout reaches 0.9609 because zero fillers implicitly suppress the spurious peak; Lemma R1(d) states the condition. Under the codebook partition of Section 4.5, neither a tag nor a position rule is needed at all. What remains is an ENCODING-LEVEL alphabet convention, not a run-time annotation.

· **Exactly one bit is a worst-case optimum, not a uniform one.** Zero-bit models reach 1.0000 on dense and 0.9949 on padded layouts, because zero fillers implicitly suppress the spurious peak. Lemma R3 proves they cannot be correct on ALL legal inputs; it does NOT claim they fail everywhere. The 2-bit result (0.2711) is strictly worse than the 1-bit result, so 'more role information is safer' is also false.

· **Not a SOTA model.** Positioning: an input-side information bound with constructive counterexample + auditable negative results. We decline every superiority claim; the withdrawn-claims register (Appendix F) is a methodological signal, not a weakness to hide.

6 Related Work, Placed Precisely

Table 1 on the first page gives the exact increment per known result; we do not repeat it here. Zoology [13], Repeat After Me [14] and Illusion of State [15] established that fixed-state models are weak at associative recall. S4 [1] / S5 [2] gave the LTI duality; Mamba [4] observed that selective parameters break the convolution view; linear attention [8] and its gated [6] / delta [7] variants gave the outer-product state. Our increment is a FORMAL tightening: a tight log2|A|-bit input-side lower bound with the bits FORCED to the input by Toeplitz duality (Theorem A), two sharp bounds with four explicit premises each, and a retraction of our own earlier mutual-exclusivity claim. We do not claim to be the first to observe that static convolutions struggle with recall, nor that input format can carry task-relevant information.

7 Limitations and a Falsifiable Prediction

The construction assumes distinct keys, adjacent binding emission, an external query, and last-layer read-out. Repeated keys, multi-query chains and stop/select mechanisms are outside T2. Finite-precision results cover L <= 128 for the analytic construction; the earlier long-decay frontier is a separate measurement. We did not run a >=10M-token equal-compute language-model experiment, and we do not present this as an architecture paper.

A falsifiable prediction the community can attack. Theorem A(b) is distribution-free and depth-free: NO deterministic decoder supplied with B < log2|A| input-side bits can be exact on all compatible instances, regardless of layer count, state order or kernel family. Any construction that achieves exact H-hop retrieval with fewer bits falsifies the bound at a precisely identifiable place. This is a target for others to test, not a request to forgive a missing experiment.

What we deliberately did NOT do. We did not add a >=10M-token equal-compute comparison, same-run Mamba-2 / GLA / DeltaNet / Hyena / Transformer baselines, or a throughput comparison. An unfair large-scale baseline would be worse than none for a theory paper. Small, clean, seeded comparisons plus an explicit 'no LM claim' are the honest scope here.

8 Conclusion

Exact retrieval needs information that a flat symbol sequence does not contain. We proved a tight lower bound on that information (B >= log2|A| bits, Theorem A), proved that Toeplitz duality FORCES the bits to the input side, showed three carriers are information-equivalent, and established that the bit-depth frontier is a right angle: extra layers cannot substitute for missing bits. Two tight structural bounds (an order-2 state for one hop; exactly H static-LTI layers for H hops under last-layer read-out) are attained by the same Toeplitz, O(1)-state construction. The practical rule is simple: budget the bits at the input or relax exactness. This is an information-bound paper with a constructive counterexample, not a claim of language-model parity or a new SOTA sequence model.

Broader Impact

This work is theoretical and empirical on synthetic tasks. It involves no human subjects, personal data, deployed systems or dual-use capabilities. Its positive potential is a quantitative design rule for constant-state models: budget log2|A| bits of role information at the input, or relax exactness. This can discourage over-engineering attention-free architectures for tasks that structurally require content addressing. Its main risk is interpretive: a reader may take 'no significant difference detected' on one synthetic task as general language-model parity, or read the supporting ledger as a list of independent discoveries. We have therefore stated every premise, sharpened the contribution to one main theorem plus two tight bounds, listed the missing large-scale experiments, and declined every superiority claim. Generative AI tools were used for implementation assistance and language editing; all research questions, conjectures and conclusions were specified by the human author, who takes full responsibility.

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Appendix A Analytic Proofs

A.0 Theorem A - A tight input-side information bound (FULL PROOF)

Status. MAIN THEOREM. Parts (a),(b) are universal information theory and hold for ANY deterministic model; we do not claim they are new. The architectural increment is (c),(d),(e),(f). Verifier: 18/18 checks (witness agreement only).

Premises P1-P5. P1 (task). An instance consists of n binding pairs, each pair being two tokens carrying symbols (u_p, w_p) from a shared alphabet Sigma of size V_K. A role assignment rho in (0,1)^(2n) decides, for each pair, which token is the KEY and which is the VALUE; together with the pair positions this defines a DIRECTED binding graph G. Pair order is arbitrary, an arbitrary number of zero FILLER tokens may appear between pairs, and symbols may repeat. The answer is k_H(G), the endpoint of the H-hop chain starting from an externally supplied query q = k_0. P2 (visible input). The model sees Vis = (x_1..x_L, s_1..s_L, q), where x_t is the SYMBOL ID ONLY (flat alphabet: keys and values share ids) and s_t in (0,1)^(b_t) is an optional side-channel; B = sum_t b_t is its total information in bits. The role assignment rho is NOT in x. The input embedding e: Sigma -> R^V is fixed and independent of t and of other tokens. P3 (model class of interest, M). H_L layers; each layer applies a STATIC LTI kernel K_h(d) = sum_p w_hp rho_hp^d (time-invariant, content-independent coefficients); fixed-dimensional state independent of L; fixed linear layer interfaces; last-layer read-out only; deterministic. NOTE: parts (a),(b) below hold for ANY deterministic model and need NONE of P3. P4 (no stop/select). Only the last layer's output is decoded; intermediate outputs are not retained. P5 (uniform instance distribution). For the accuracy statement, instances sharing the same Vis are equiprobable.

Claim (a)-(f). (a) UNIVERSAL IMPOSSIBILITY. For any deterministic model (no LTI, depth, state or read-out assumption), if |A(x,q,H)| >= 2 then the model cannot be correct on all instances compatible with Vis; accuracy <= 1/|A| <= 1/2. (b) BIT LOWER BOUND. Exact retrieval on all compatible instances requires B >= log2 |A(x,q,H)| bits. There is an explicit family with |A| = 2^m for every m <= n, hence B >= m bits; taking m = n gives B >= n. If the pairing itself is also not encoded in x (fully non-adjacent binding), |A| >= n! and B >= log2(n!) = Theta(n log n). The bound is a property of the TASK, independent of H_L, state order P, kernel parameters (rho, theta, w), and the number of layers. (c) THE CLASS M CANNOT MANUFACTURE THESE BITS. Within P3, increasing H_L, P, or changing (rho, theta, w) cannot lower the bound. The only content dependence of the computation enters through e(x_t) and the absolute position t; under arbitrary filler spacing the pair positions are independent of rho, and e is fixed, so the output is invariant under changes of rho that leave Vis fixed. To make the kernel itself carry role information one must make its coefficients input-dependent and hence time-varying; by T4 (Toeplitz IFF parameter-time-invariant for the second-order recurrence with c_t != 0), that destroys the Toeplitz structure and the single-FFT / IIR duality. Therefore: TO KEEP DUALITY, THE MISSING BITS MUST BE SUPPLIED ON THE INPUT SIDE. (d) TIGHTNESS AND CONSERVATION. The bound is attained by three information-equivalent carriers, each supplying exactly 1 bit per ambiguous token: (i) an explicit 1-bit KEY/VALUE tag; (ii) an alphabet partition V -> 2V (Corollary R2), which is precisely log2(2) = 1 bit per token folded into the id and PRESERVES Toeplitz, O(1) state and zero-training exactness; (iii) an absolute-position format prior (the withdrawn W9), which encodes the same bit in an ASSUMPTION ABOUT THE INPUT DISTRIBUTION rather than in the data. When arbitrary filler spacing is allowed the assumption in (iii) ceases to hold, its effective information drops to zero, and accuracy collapses quantitatively (1.0000 -> 0.3438 +/- 0.0314). Information does not disappear; it only changes carrier, and no carrier can beat (b). (e) THE BIT-DEPTH FRONTIER IS A RIGHT ANGLE. For every B >= log2|A| the minimum required depth is still exactly H (Theorem 2); for every B < log2|A| NO depth suffices. There is NO smooth rate at which extra layers buy fewer input bits. Consequently 'b bits <-> minimum layers' is a step function, not a trade-off curve. (f) OVERSHOOT IS NOT AN ARCHITECTURAL LAW. The failure at H_L > H is a consequence of P4 (last-layer read-out, no stop/select). Retaining intermediate outputs and selecting the H-th makes H_L >= H sufficient; the LOWER bound H is unchanged. Necessity of exactly H layers must not be read as a law of the architecture.

Proof (a). The output y = F(Vis) is a single deterministic value. Two instances G, G' in R(x) with the same Vis require two distinct answers k_H(G) != k_H(G'). F(Vis) can equal at most one of them. Under P5 the fraction correct is at most 1/|A| <= 1/2. No property of F beyond determinism is used; in particular no assumption about kernels, state order, depth or read-out. QED.

Proof (b). The side-channel is a function S: R(x) -> (0,1)^B. F may depend on S, so instances can be distinguished only if S is injective on the answer classes of R(x): whenever k_H(G) != k_H(G') we need S(G) != S(G'), otherwise (a) applies to the pair. A map into (0,1)^B has at most 2^B fibres, so 2^B >= |A|, i.e. B >= log2|A|. EXPLICIT FAMILY WITH |A| = 2^m (binary-chain construction). Fix m <= n and take H = m hops. At hop h the visible sequence contains one key symbol k_h and TWO candidate value symbols p_h, q_h; a bit d_h in (0,1) that is NOT recorded in x selects which candidate is the valid binding, and the selected value is the key of hop h+1. The two candidates are both present as tokens, so every vector d = (d_1..d_m) yields the SAME visible symbol sequence x and the same query q = k_1. Map d to the endpoint id sum_h d_h 2^(h-1); with V >= 2^m these are 2^m DISTINCT endpoints, hence |A(x,q,H)| = 2^m and B >= m. The witness enumerates all 2^m assignments and confirms bit-identical visible inputs (max deviation 0.0). Choosing m = n gives B >= n. FULLY NON-ADJACENT CASE. If even the pairing is not recoverable from x, R(x) contains all n! matchings between n key slots and n value slots; by the same injectivity argument B >= log2(n!). Because the main theorem already holds in the STRICTER (more model-friendly) setting where pairing IS recoverable from x, it holds a fortiori when it is not. QED.

Proof (c). In M, layer h computes z_h = sum_t K_h(L-1-t) m_h[t] b_t with m_h[t] = <q_h, e(x_(t-1))> and b_t = e(x_t), and q_(h+1) = T_h(z_h) with T_h fixed linear. K_h depends only on the lag d and on fixed parameters, never on x; T_h is fixed; e is fixed. Therefore the entire composition is a function of the multiset of (symbol id, position) pairs plus q, i.e. of Vis with s removed. Under arbitrary filler spacing the positions of the pair tokens are determined by filler counts that are independent of rho, so the mapping rho -> (symbol, position) data is constant on R(x). Hence F is constant on R(x): no choice of H_L, P or (rho,theta,w) changes this, and (a) applies with full force. Now suppose one tries to encode rho in the KERNEL instead. Then the recurrence coefficients become functions of x_t and, since x varies with t, they become time-varying. For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, T4 proves the impulse response is Toeplitz IFF (a,b,c) are constant in t; time-varying coefficients give a non-Toeplitz h(t,s), no single H(z), and the single-FFT path fails (we negate THAT path, not every conceivable O(L log L) algorithm). So within the architecture family that keeps duality, the bits can only enter on the input side. QED.

Proof (d). TIGHTNESS. Carrier (i): set s_t = 1 bit equal to the KEY/VALUE role; then S(rho) = rho is injective on answer classes and B = 2n >= n, so (b) is attained (up to the constant factor 2 from tagging both tokens of each pair; tagging only the key token gives B = n exactly, which is tight). Carrier (ii): R2 gives an exact construction with no run-time tag by using two orthogonal halves of the codebook; the id of each token now carries log2(2V_K / V_K) = 1 bit, and 1.0000 accuracy is attained on all four layouts with zero training, Toeplitz deviation 0.0 and FFT/IIR error 2.89e-16. So the bit is present, just folded into the alphabet. CONSERVATION. All three carriers supply the same 1 bit per ambiguous token: (i) as explicit data, (ii) as alphabet capacity, (iii) as a distributional assumption. Carrier (iii) is free ONLY while the assumption holds. The W9 experiment breaks exactly that assumption (random zero fillers make key parity independent of the answer); the effective supplied information falls to 0, the bound (b) is violated, and accuracy falls to 0.3438 +/- 0.0314 (5 seeds), consistent with (a)'s <= 1/2 prediction rather than with chance 1/V_K. QED.

Proof (e). By (b), B < log2|A| makes exactness impossible for EVERY deterministic model, hence for every H_L; increasing depth cannot help. By Theorem 2 (NT31/R2), when B >= log2|A| the minimum depth is exactly H, with H_L < H at chance and H_L > H overshooting under P4.

The frontier in (B, minimum H_L) is therefore ((B < log2|A|): infeasible) union ((B >= log2|A|): H_L = H), a right angle. QED.

Proof (f). Under P4 the decoded value at depth H_L is k_(H_L). For H_L > H this differs from k_H because chain keys are distinct. If instead the H-th intermediate output is retained and selected, the answer is k_H for any H_L >= H; the lower bound H_L >= H is unchanged (a single layer still cannot perform two content matches). Hence 'exactly H' is a statement about the read-out convention, not an architectural law. QED.

NON-VACUITY. n = 4, m = 2, V_K = 32, H = 2, H_L = 2, flat alphabet, arbitrary filler spacing: B = 0 gives |A| = 4 and accuracy <= 1/4; B = 2 (tag only the two key tokens) gives 1.0000; the empirical witnesses in raw result files in the supplementary artifact trace the step at B = m.

Boundary / scope. SCOPE. (i) Parts (a),(b) are UNIVERSAL and make NO architectural assumption; we deliberately do NOT claim they are new information theory. The novelty is (c),(d),(e): the FORCED LOCATION of the bits under Toeplitz duality, three-way conservation with a tight duality-preserving construction, and the right-angle frontier. (ii) 'Exact' means correct on EVERY compatible instance, not high average accuracy; a probabilistic or majority decoder is outside P5. (iii) The bit bound is on INPUT-SIDE information. It does NOT say a model cannot learn a role predictor from a training distribution; it says that when the visible input genuinely lacks the information, no deterministic decoder can recover it. A learned prior that guesses rho from x is exactly carrier (iii) and inherits its fragility. (iv) Constant factors: tagging both tokens of a pair costs 2n bits while the tight value is n; we state the tight version. (v) The |A| = 2^m family requires V_K >= 2^m; for arbitrary V_K take m <= floor(log2 V_K) or use R = 2^ceil(log2 V_K) Hadamard rows. (vi) (c) negates the single-FFT convolution path, not every possible O(L log L) algorithm, and is proved for the second-order recurrence with c_t != 0 (T4's scope). (vii) The theorem is about SYNTHETIC content-chain retrieval. It makes no claim about language modeling, perplexity, generation, throughput, or superiority over any baseline. (viii) RELATION TO LEMMA R3. An earlier lemma in this line of work states that 'exactly one role bit is necessary'. That statement is about a PER-TOKEN symbol-role distinction and is the WORST-CASE instance |A| = 2 of Theorem A: R3 exhibits one legal dataset on which zero bits force every model in the class to output the distractor (accuracy 0), whereas (a) gives the weaker uniform-average bound 1/|A| = 1/2. Theorem A does NOT overturn R3; it REFINES it in two ways: the necessary quantity is the TOTAL input-side information B >= log2|A| (not 'exactly one bit'), and |A| can be 2^m for any m <= n, so the requirement grows with the number of distinct compatible answers. Consequently 'one bit per token' is a SUFFICIENT carrier, not a necessary one, and an explicit role tag is neither necessary nor the information-theoretically minimal carrier. (ix) 'Side-channel' includes any deterministic function of the instance that the model can observe; folding it into token ids, positions or an explicit tag are all counted in B.

Novelty, stated precisely. Zoology (2312.04927), Repeat After Me (2402.01032) and Illusion of State (2404.08819) establish that fixed-state models are weak at associative recall; S5 (2208.04933) and Mamba (2312.00752) establish that LTI requires time-invariant coefficients and that selectivity breaks the convolution view. None of these quantifies a BIT lower bound on input-side role information, none proves that the bits are FORCED to the input side by Toeplitz duality, and none identifies the right-angle bit-depth frontier. Our increment is the exact quantification plus the location theorem plus conservation; we do NOT claim to be the first to observe that input format can carry task-relevant information. Our own earlier one-bit lemma (R3) is the |A|=2 worst-case special case; Theorem A generalizes it to a quantitative total-bit bound and removes the 'exactly one bit' wording, which was an over-statement of necessity.

Architectural use. Converts the three killer objections into answers stated IN THE MAIN TEXT, before rebuttal. 'The format solves the task' -> yes, and Theorem A says exactly how many bits it must supply and proves the format cannot be beaten. 'Isn't this linear attention with extra steps' -> (c) shows that keeping Toeplitz (which linear attention with rho=1 also does) forces the bits to the input; the separation is about WHERE the information lives, not about the state update. 'Toy synthetic, zero training' -> (b),(e) are distribution-free and depth-free, so they are not toy facts that more layers would fix. Engineering rule: budget at least log2|A| bits on the input side, or relax exactness; extra layers cannot substitute.

Witnesses (not the proof). Analytic proof above. Empirical witnesses (NOT the proof): raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact. Code: the supplementary artifact Related: NT31/R2 (tight construction), T4/NT12 (Toeplitz IFF), W9 (format prior).

All 31 proofs are analytic. The main paper promotes SIX of them to core theorems (T1-T6) and demotes eight standard facts to Propositions P1-P8 in Appendix B; the full ledger is retained here as an auditable evidence chain, NOT as a claim of 31 independent discoveries. Finite-identity checks certify statement formation and non-empty premises only; they are not part of any proof. Symbol convention: t = time index, tau = lag, d = t - s, H = hops, H_L = layers.

A.31 NT31 - Exact-H layer tightness for content-chain retrieval

Premises. H1. A sequence of n ordered pairs (k_p, v_p), p=0..n-1, is presented as a strictly alternating token stream x_0=k_(pi(0)), x_1=v_(pi(0)), ... under an arbitrary permutation pi of pair positions; all keys k_p are DISTINCT and all values v_p=k_(p+1) lie in the key set (k_0..k_n) (a content chain). H2. The query q=k_0 is placed at the last position t=L-1, L=2n+1. H3. A fixed deterministic code C in R^(V x R) satisfies <C_i,C_j>=delta_ij for i,j in the key range (orthogonality is SUFFICIENT; the proof only needs the self-inner-product to strictly dominate off-diagonal ones). H4. Each of H_L layers is a static LTI tensor spectral convolution with kernel K_h(d)=sum_(p=1..P) w_(h,p) rho_(h,p)^d, w_(h,p)>0, 0<rho_(h,p)<1, so K_h(d)>0 for every d>=0. H5 (TIGHT in v6). The INPUT ENCODING provides exactly ONE BIT of key/value role distinguishability per symbol: KEY symbol k uses C[k] and VALUE symbol k uses C[V_K+k] from two ORTHOGONAL halves of the codebook. Lemma R3 proves 0 bits are insufficient for uniform correctness (an adversary pair permutation forces failure for ANY layer interface, learned or analytic); R2 proves 1 bit suffices; R2(d) proves a 2nd bit (any extra position-only gate) is STRICTLY HARMFUL. Hence the bit count is exactly one, necessary and sufficient, and endogenous to the model's own codebook. distinction; no run-time annotation, absolute-position rule or format prior is required. HISTORICAL CONVERGENCE (W9): an earlier draft used absolute-position parity r[t]=1[(t-1) even], which is a hidden format prior and was WITHDRAWN; a later draft used an external 1-bit run-time role tag. Both are stronger premises than the codebook partition and are no longer needed. H6. Layer interface q_(h+1)=z_h/||z_h||; the final layer output z_(H_L) is decoded by logits_j = <z_(H_L), C_j> over the key range. H7. 1<=H<=n, and the readout is the LAST layer (no stop/select mechanism).

Claim. (a) LOWER BOUND. For every h<=min(H_L,n), the h-th layer output is exactly z_h = c_h C[k_h] with c_h = K_h(L-1-t_h) > 0, where t_h=pos(k_(h-1))+1 is UNIQUE. Hence if H_L < H the decoded answer is k_(H_L) != k_H and accuracy is at most chance under independently re-assigned values. (b) CONSTRUCTION. If H_L = H the decoded answer is exactly k_H with ZERO training, for every n, every pair permutation, every V, and every admissible (rho,w,C). (c) OVERSHOOT. If H_L > H (and H_L<=n) the last-layer answer is k_(H_L) != k_H, so accuracy is again at most chance. More layers do NOT help without a mechanism that selects the H-th intermediate output. (d) TIGHTNESS. H_L = H is both necessary and sufficient; the total state is O(H_L P R) scalars, INDEPENDENT of L. Every kernel is static => Toeplitz => single-FFT / two-tap-IIR duality holds (NT12); no N x N score matrix, no softmax over tokens, no KV cache, strict causality. (e) ROLE BASIS IS NECESSARY. If r[t] is removed, the same token plays two roles (previous value and next key), the matching signal m_2 has TWO unit peaks, and two orthogonal codes superpose; accuracy drops strictly below 1.

Proof (a),(b). Induction on h (written for the general H5; under implementation (i) the gate r[t] is the identity by Corollary R2). Base h=1: m_1[t] = <q_1, r[t] C[x_(t-1)]> = <C[k_0], r[t] C[x_(t-1)]>. By H3 the inner product is delta_(k_0,x_(t-1)); by H5 the factor r[t] retains only positions whose predecessor carries the KEY role tag. All keys are distinct (H1), so k_0 occurs as a key exactly once, at some even index 2p; therefore m_1[t]=1 at the unique t_1=2p+1 and 0 elsewhere. At t_1 the current token

is v_(pi(p))=k_1, so z_1 = K_1(L-1-t_1) C[k_1] = c_1 C[k_1] with c_1>0 (H4). Inductive step: q_(h+1)=z_h/||z_h|| = C[k_h] because c_h>0 and ||C[k_h]||=1. Repeating the base argument with k_h in place of k_0 gives a UNIQUE match at t_(h+1)=pos(k_h)+1, hence z_(h+1)=c_(h+1) C[k_(h+1)], c_(h+1)>0. Decoding: logits_j = <c C[k_(H_L)], C_j> = c delta_(k_(H_L),j), so argmax_j logits_j = k_(H_L). For H_L<H, k_(H_L) != k_H (distinct chain) => wrong; for H_L=H => exactly k_H. QED.

Proof (c). Same induction gives z_(H_L)=c C[k_(H_L)]; with H_L>H and H_L<=n, k_(H_L)!=k_H => wrong. QED.

Proof (d). (a)+(c) give necessity, (b) sufficiency => tight. Each layer keeps P state vectors s_(h,p) in R^R updated by s_(h,p) <- rho_(h,p) s_(h,p) + u_t, plus one query vector q_h in R^R; total O(H_L P R) scalars with no L dependence. Because rho_(h,p) and w_(h,p) do not depend on t or on the input, the impulse response is a function of the delay d alone (Toeplitz), hence a single FFT realizes exactly the same operator as the IIR scan (Theorem 1 / NT12). The role basis multiplies u_t, not the kernel, so B4/NT12 is not triggered. Strict causality: u_t depends only on x_(t-1), x_t. QED.

Proof (e). Without r[t], for h=2 the query is C[k_1] and x_(t-1)=k_1 occurs at TWO positions: the odd index where k_1 is a VALUE (of the pair k_0->k_1) and the even index where k_1 is a KEY (of the pair k_1->k_2). Thus m_2 has two unit peaks and z_2 = c C[k_2] + c' C[k_wrong] with C[k_2] orthogonal to C[k_wrong]; the argmax is decided by the relative kernel weights at the two positions, which vary with the random permutation pi, so accuracy is strictly below 1. QED.

Non-vacuity: n=4, V=64, C=Sylvester Hadamard, P=2, rho=(0.90,0.999), w=(0.5,0.5), H=1..4 satisfies H1-H7 and attains 1.0 on the diagonal.

Boundary / scope. SCOPE. (i) Distinct keys and strict key/value alternation are essential; with repeated keys the uniqueness of t_h fails and de-duplication is NOT proved here. (ii) 'Exactly H' is under LAST-LAYER readout with no stop mechanism; if intermediate outputs are retained and the H-th is selected, then H_L>=H also suffices (the lower bound H remains, the upper bound disappears). (iii) Orthogonality of C is sufficient but not necessary; only self-inner-product dominance is needed, which is why a Gaussian i.i.d. code also attains 1.0 in the ablation. (iv) Layer normalization q=z/||z|| is a KERNEL-EXTERNAL nonlinearity: rho and theta remain time-invariant, so B4 does not apply. (v) PREMISE RELAXATION (W9 then R2). An earlier draft used the absolute-position parity mask r[t]=1[(t-1) even] as H5. That is a HIDDEN FORMAT PRIOR, sufficient ONLY when keys always sit at even positions; with arbitrary filler spacing it drops from 1.0 to 0.3438 +/- 0.0314 (W9). H5 was then a 1-bit run-time role tag, and is now RELAXED again: Corollary R2 shows that an orthogonal codebook partition needs NO gate at all and still attains the same tight bound on all four layouts. Adding a parity gate back is strictly HARMFUL (R2(d)). All three implementations act on the INPUT SIGNAL, never on the kernel, so Toeplitz / FFT-IIR duality are preserved. (vi) RELATION TO LINEAR ATTENTION: at rho=1 with a single pole, z_h = q_h^T (sum_t r[t] C[x_(t-1)] C[x_t]^T), which is isomorphic to a linear-attention state; with rho<1 and multiple poles K_h(d)=sum_p w_p rho_p^d is a DECAY-SPECTRUM weighted version. We do NOT claim to leave that family. The contribution is the joint tightness of (order 2, exactly H layers, O(1) state, static Toeplitz kernel, no N x N) -- not a claim of a new paradigm. (vii) This is a SYNTHETIC content-chain retrieval theorem. It does NOT establish language-model parity, perplexity, or generation quality.

Architectural use. Closes the architecture-search corner left open by NT11/NT28: multi-hop content addressing is compatible with FFT/IIR duality after all, at the exact price of one static-LTI layer per hop plus a theta->pi role basis. Predicts three falsifiable observations: single-layer H>=2 must be at chance; H layers must be exactly 1.0 with ZERO training; H+1 layers must FAIL. It also converts the earlier '0.5918 partial construction' into a tight construction and explains the residual as ROLE AMBIGUITY, not capacity.

Witnesses (not the proof). Analytic proof above. Empirical witnesses (NOT the proof): raw result files in the supplementary artifact (5x5 layer x hop grid), raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact (5 seeds); raw result files in the supplementary artifact, raw result files in the supplementary artifact (W9 correction); raw result files in the supplementary artifact, raw result files in the supplementary artifact (R2); raw result files in the supplementary artifact, raw result files in the supplementary artifact (R3 necessity, v6 tightness). Verifier: 17/17 checks pass.

A.32 Lemma R1 - Minimality and irreducibility of the one-bit role tag

Status. SUPPORTING LEMMA, not a promoted core theorem. Hypotheses P1-P4 are stated in full below; the proof is analytic. Verifier: 12/12 checks (witness agreement only).

Premises. P1. A content chain of n pairs (k_p, k_(p+1)) with DISTINCT keys; each pair is presented as two ADJACENT tokens (key immediately followed by value); pair order is arbitrary; between pairs any number of zero FILLER tokens may be inserted. P2. Static LTI kernel K(d)=sum_p w_p rho_p^d > 0, fixed deterministic code C with C C^T = I on the key range, query supplied externally. P3. The layer interface is any FIXED (non-learned, content-independent) linear map followed by normalization; all coefficients of the kernel are time-invariant. P4. Available side information per position t is a binary gate w_t in (0,1); w_t may depend on absolute position t, or on a role tag attached to the input symbol, but NOT on the answer k_H or on the pair permutation in a way that leaks it.

Claim. (a) POSITION-ONLY GATES ARE FORMAT PRIORS, NOT ARCHITECTURE. Let w_t = 1[(t-1) even] (the v5 parity mask). If each pair is preceded by f_p filler tokens drawn independently with P(f_p odd) >= 1/2, then for every hop h the indicator w at the unique key-successor position is 1 with probability at most 1/2 and is INDEPENDENT of k_h. Whenever it is 0 the true peak is suppressed and z_h carries no information about k_h; hence expected accuracy <= 1/2 + 1/(2 V_K). (b) ONE BIT IS SUFFICIENT AND FORMAT-INDEPENDENT. Let w_t = 1[role(x_(t-1)) = KEY], where each input symbol carries a deterministic KEY / VALUE / FILLER tag. Then for every pair permutation and every filler pattern the NT31 induction gives z_h = c_h C[k_h] with c_h>0, so accuracy is exactly 1 with zero training. (c) CODE-SUBSPACE PARTITION CANNOT REPLACE THE BIT. Suppose we split C into two orthogonal row subspaces C_key, C_val, set A_t = C_key[x_(t-1)], b_t = C_val[x_t], w_t == 1 (no role information), and use any FIXED linear layer interface T. Then accuracy is strictly below 1 on the tight alternating layout. (d) NECESSITY IS LAYOUT-CONDITIONAL. With w_t == 1 and a TIGHT alternating layout, accuracy is below 1. With arbitrary filler spacing, filler tokens implicitly zero the spurious peak and accuracy can exceed 0.95. Therefore 'role information is necessary' must NOT be stated as an unconditional theorem. (e) The role tag acts on the INPUT SIGNAL only; every kernel remains static and Toeplitz, so FFT/IIR duality and O(1) state are preserved.

Proof (a). The p-th key sits at absolute position s_p = sum_(q<p)(f_q+2) + f_p. Its successor is t_p = s_p + 1, and w_(t_p) = 1[(s_p) even]. Because f_p enters s_p additively and P(f_p odd) >= 1/2, the parity of s_p is independent of the pair permutation and of the answer k_h, and P(w_(t_p)=1) <= 1/2. Condition on the event w_(t_p)=0 (probability >=1/2): the h-th matching signal m_h[t_p]=0, so by NT31's base step with distinct keys the only remaining contributions are either zero (filler predecessors) or value-subspace codes orthogonal to q_h; in either case z_h is independent of k_h and the decoded answer is uniform on V_K, giving accuracy 1/V_K. Unconditionally, accuracy <= 1/2*1 + 1/2*(1/V_K) <= 1/2 + 1/(2 V_K). QED.

Proof (b). Replace every occurrence of 'even index' in the NT31 induction by 'role(x_(t-1)) = KEY'. By P1 a KEY tag appears exactly once per pair, and by distinctness k_(h-1) tags exactly one pair, so the match is unique for ANY filler pattern. All other steps are verbatim. QED.

Proof (c). In the tight alternating layout the symbol k_h occurs TWICE: as the VALUE of pair h-1 and as the KEY of pair h. Hence A_t = C_key[k_h] at two distinct positions t_a, t_b. Let T be any fixed linear layer interface and q_(h+1) = T(z_h). Case 1: q_(h+1) has zero inner product with C_key[k_h]; then m_(h+1) == 0 at both positions, z_(h+1)=0, and accuracy is chance. Case 2: the inner product is nonzero; then BOTH positions contribute, and z_(h+1) = K(t_a) C_val[w] + K(t_b) C_val[k_(h+1)], where w is the key of the pair adjacent to pair h-1 in the RANDOM permutation and is independent of k_(h+1). Because C_val[w] and C_val[k_(h+1)] are

orthogonal, argmax selects k_(h+1) iff K(t_b) > K(t_a), an event determined by positions, not by the answer; over random permutations this fails with strictly positive probability. Hence accuracy < 1 for every fixed T. QED.

Proof (d). Tight layout gives two nonzero peaks => (c) applies. In a gapped layout the token following a VALUE is a zero FILLER, whose b_t is the zero vector, so the spurious peak contributes nothing and only the true peak survives; accuracy can approach 1. Thus the necessity of the bit is a property of the LAYOUT, not a universal law. QED.

Proof (e). w_t multiplies A_t = C[x_(t-1)], i.e. the input signal u_t. The impulse response K(d) is unchanged and depends only on the delay; Toeplitz holds, so a single FFT realizes the same operator as the IIR scan. The state is P vectors per layer, independent of L. QED.

NON-VACUITY: n=4, V=64, VK=32, C = Sylvester Hadamard, f_p ~ Uniform(0..4), H=2, H_L=2 satisfies P1-P4; (a) witnesses 0.3438 +/- 0.0314 (5 seeds), (b) witnesses 1.0000 on all four layouts, (c) witnesses 0.8691, (d) witnesses 0.8621 (none, tight) vs 0.9609 (none, gapped).

Boundary / scope. SCOPE. (i) This is a SUPPORTING LEMMA, not a 32nd headline theorem; the v5 main paper promotes exactly six core theorems. (ii) The upper bound in (a) is an expectation over the filler distribution, not a worst-case bound for every fixed filler pattern; a pattern with all keys at odd positions would give exactly chance, while one with all keys even gives 1. (iii) The 1-bit tag is minimal among SIDE-CHANNELS but we do NOT prove that no architecture outside P1-P4 (e.g. learned input-dependent gates, extra recurrence, or retaining intermediate outputs) could work without it. THIS RESERVATION HAS NOW BEEN EXERCISED: Corollary R2 changes P1 itself by splitting the input alphabet into two orthogonal codebook halves, so key and value symbols are DIFFERENT ids. R2 attains 1.0000 with NO side-channel at all. R1(c) is therefore NOT overturned -- it remains valid for the flat encoding it assumes -- but its conclusion must NOT be read as 'no architecture can work without a role tag'. (iv) 'KEY / VALUE / FILLER' is a SYMBOL-TYPE annotation, not an answer leak: it does not reveal which key is queried or what k_H is. (v) Filler tokens are zero one-hots; their presence is itself structural information, which is precisely why (d) refuses an unconditional necessity claim. (vi) Synthetic content-chain task only; no LM claim. (vii) WITNESS CAVEAT: the first implementation of the (c) candidate used (A_prev @ Ck.T @ Ck), which is the projection Ck^T Ck rather than the key code. That bug was found and fixed; the recorded 0.8691 is from the corrected implementation and equals the no-role baseline, confirming the subspace partition adds nothing.

Architectural use. Converts the most damaging potential rejection ('the construction hides a task-format prior') into an auditable scope correction. Justifies withdrawal W9 and the H5 revision of NT31. Gives a concrete engineering rule: emit a role tag with each symbol (or use separate key/value symbol ids) instead of relying on absolute position parity. Also explains why the earlier 0.5918 two-layer result and the 0.8691 no-role baseline share the same root cause: role ambiguity, not capacity.

Witnesses (not the proof). Analytic proof above. Empirical witnesses (NOT the proof): raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact. Code: the supplementary artifact R2 (which exercises SCOPE (iii)): the supplementary artifact

A.33 Corollary R2 - Codebook-intrinsic role separation

Status. COROLLARY of NT31 (premise relaxation), NOT a new headline theorem and NOT a new ledger item. Derived from: NT31 (premise relaxation) + NT04/NT05 (orthogonal code). Verifier: 17/17 checks (witness agreement only).

Premises Q1-Q5. Q1. C in R^(V x V) is a Sylvester Hadamard matrix with V = 2 V_K a power of two, so C C^T = I_V and <C[i],C[j]> = delta_ij. Q2. The input alphabet has THREE disjoint symbol classes: KEY symbols with ids 0..V_K-1, VALUE symbols with ids V_K..2V_K-1, and a zero FILLER. A pair (k_p, k_(p+1)) is emitted as the adjacent KEY symbol k_p followed by the VALUE symbol V_K + k_(p+1); filler tokens may appear between pairs; pair order is arbitrary. This is an ENCODING CONVENTION, not an annotation attached at run time. Q3. All keys k_0..k_n are DISTINCT; query q_1 = C[k_0] is supplied externally; the answer k_H is decoded in the KEY half. Q4. H_L layers, each with static kernel K_h(d)=sum_p w_hp rho_hp^d > 0; fixed layer interface T = C_key^T C_val (an orthogonal change of basis inside the codebook, content-independent and with no learned weights). Q5. LAST-layer readout only, no stop mechanism, 1 <= H <= n.

Claim. (a) NO GATE IS NEEDED. For every hop h, the matching signal m_h[t] = <q_h, C[x_(t-1)]> has exactly ONE nonzero peak, at the position whose predecessor is the KEY symbol k_(h-1). The value occurrence of the same chain element contributes ZERO because <C[k], C[V_K + j]> = 0 for all k, j. (b) INDUCTION. z_h = c_h C[V_K + k_h] with c_h = K_h(L-1-t_h) > 0, and q_(h+1) = (z_h @ C_val^T) @ C_key = C[k_h]. (c) TIGHTNESS PRESERVED. H_L = H gives exactly k_H with zero training, for every n, pair permutation, filler pattern and V; H_L < H fails (lower bound) and H_L > H OVERSHOOTS and fails. (d) EXTRA GATES ARE HARMFUL, NOT HARMLESS. Composing any position-only gate g[t] with the codebook encoding can only REMOVE the unique true peak. If g[t_p] = 0 at the key-successor position (which occurs with positive probability under arbitrary filler spacing), then m_h == 0, z_h = 0, and accuracy is chance. The parity gate is therefore strictly worse than no gate. (e) CONSERVATION. Every kernel remains static and Toeplitz (FFT/IIR duality), strict causality holds, state is O(H_L P V) scalars independent of L, no N x N score matrix, no softmax over tokens, no KV cache, NO external role tag and NO absolute-position gate. (f) PREMISE ORDERING. Among sufficient implementations of NT31 H5, codebook partition is the WEAKEST: it requires no side-channel and no format assumption. An external 1-bit tag is a strictly stronger implementation; absolute-position parity is sufficient ONLY under a strict alternating format (W9).

Proof (a). By Q1, C's rows are mutually orthonormal. Write the predecessor symbol at position t as either a KEY code C[k], a VALUE code C[V_K + j], or the zero vector. q_h = C[k_(h-1)] by (b)'s induction hypothesis (base q_1 = C[k_0] by Q3). Then <C[k_(h-1)], C[V_K + j]> = 0 for every j, and <C[k_(h-1)], 0> = 0. Hence m_h[t] is nonzero only when the predecessor is the KEY symbol k_(h-1); by Q3 distinctness there is exactly ONE such position t_h. No masking, no tag, no position information is used. QED.

Proof (b). m_h[t_h] = 1 and b[t_h] = C[V_K + k_h] by Q2, hence z_h = K_h(L-1-t_h) C[V_K + k_h] with positive coefficient by Q4. Because C_val has orthonormal rows, z_h @ C_val^T = c_h e_(k_h) (a one-hot coordinate vector), and e_(k_h) @ C_key = C[k_h]. Thus q_(h+1) = C[k_h] after normalization (c_h > 0). QED.

Proof (c). Decoding logits_j = <z_H, C[V_K + j]> = c_H delta_(k_H, j), so argmax = k_H iff H_L = H. For H_L < H the last layer outputs k_(H_L); for H_L > H it outputs k_(H_L); both differ from k_H because the chain keys are distinct. Filler tokens contribute zero to z_h by Q2, so the argument is independent of the filler pattern. QED.

Proof (d). Let g[t] in (0,1) depend only on t. The gated matching signal is g[t] m_h[t]. By (a), m_h has a single nonzero peak at t_h, so the gated signal is identically zero whenever g[t_h] = 0, and then z_h = 0 for every h >= that layer. Under arbitrary filler spacing the parity of t_h - 1 is not fixed by the answer, so P(g[t_h] = 0) > 0. Unconditionally accuracy <= (1 - P) * 1 + P * (1/V_K) < 1. The ungated construction attains 1. Hence the extra gate is strictly harmful. QED.

Proof (e). Q4 fixes (rho_hp, w_hp) in t and in the input, so the impulse response depends only on the delay d (Toeplitz); a single FFT realizes the same operator as the IIR scan. T is a fixed linear map applied between layers, outside the recurrence. Each layer keeps P state vectors of dimension V plus one query vector, so the total is O(H_L P V) with no L dependence. Causality holds because z_h uses only positions <= L-1. QED.

Proof (f). Compare the three implementations of NT31 H5: parity needs the extra assumption 'all KEY symbols sit at even positions' (W9 shows it fails otherwise); the 1-bit tag needs an EXTERNAL per-symbol annotation; the codebook partition needs only an alphabet convention that is part of the model's own codebook. The set of admissible inputs is therefore strictly larger under Q2 than under either alternative, while the conclusion (c) is the same. QED.

NON-VACUITY. V=64, V_K=32, n=4, H=2, H_L=2, C=Sylvester

Hadamard, P=2, rho=(0.90,0.999), w=(0.5,0.5) satisfies Q1-Q5 and attains 1.0000 on all four layouts (alternating, gapped, dense, padded) with zero training; the 5x5 grid is 1.0000 on the diagonal and <=0.10 off.

Boundary / scope. SCOPE. (i) R2 is a COROLLARY of NT31, NOT a 32nd headline theorem and NOT a new ledger item; the 51-item ledger and the six core theorems are unchanged. (ii) Lemma R1 is NOT overturned: R1 assumes P1 in which both pair tokens share ONE id space. R2 changes that INPUT ENCODING, which is precisely the escape R1 SCOPE (iii) reserves. R1(c)'s impossibility holds for fixed linear layer interfaces UNDER the flat encoding, and R2 uses the same fixed interface under a DIFFERENT encoding. (iii) The alphabet convention is still information: a KEY symbol and a VALUE symbol are different ids. We claim NO EXTERNAL SIDE-CHANNEL, not 'no structural convention whatsoever'. A reviewer who insists that any key/value distinction is a task prior is correct that this is an encoding-level inductive bias; we state it openly. What is removed is the RUN-TIME annotation, the absolute-position assumption and the format prior. (iv) Distinct keys, adjacent (key,value) emission, external query and last-layer readout remain premises. Repeated keys degrade accuracy (witness 0.9414). (v) V must be a power of two for the Sylvester construction; for arbitrary V take R = 2^ceil(log2 V) rows (NT04/NT05, R < 2V). (vi) A dense V x V codebook is never stored: the fast Walsh-Hadamard transform applies C in O(V log V) time and O(V) memory, so V=2^18 (Qwen3 scale, explicit matrix 549.8 GB) takes about 0.010 s per vector. This is the same structural advantage as the FFT path. (vii) Finite-precision results cover L<=128 only; the output is a rank-1 sparse direction, so quantization cannot change argmax. This does NOT extend the earlier long-decay precision frontier. (viii) Synthetic content-chain task; no LM perplexity, no generation, no language-model parity claim.

Architectural use. Closes the last reviewer objection left by W9: 'an explicit KEY/VALUE tag is still an external annotation'. It converts the role distinction from a run-time label into an intrinsic property of the spectral-convolution layer's own orthogonal codebook, and proves that adding a role gate back is HARMFUL. It also gives a concrete engineering rule for real vocabularies: split the codebook into two orthogonal halves and use FWHT instead of materializing C. The three-step H5 relaxation (parity -> tag -> codebook) is a monotone weakening of assumptions, not a sequence of reversals.

Witnesses (not the proof). Analytic proof above. Empirical witnesses (NOT the proof): raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact. Code: the supplementary artifact

A.34 Lemma R3 - zero role bits are insufficient

Status. Necessity half of NT31's tightness; NOT a new headline theorem and NOT a new ledger item. Verifier: 23/23 checks.

Premises P1-P4. P1 (FLAT alphabet, zero role bits). Every symbol id k has a single code C[k]; KEY symbols and VALUE symbols share it. C C^T = I_V on the id range. Filler tokens carry the zero code. No run-time role tag, no absolute-position gate, no pair-boundary rule. P2 (model class = NT31/R2). H_L = H static-LTI layers. Each layer keeps an order-2 state S_h = sum_t K_h(t) A_t (x) b_t with A_t = Phi_A C[x_(t-1)], b_t = Phi_b C[x_t] for FIXED linear Phi_A, Phi_b; K_h(t) = sum_p w_hp rho_hp^(L-1-t) with 0 < rho_hp < 1, hence K_h(t) is strictly increasing in t. Read-out z_h = S_h^T q_h; layer interface q_(h+1) = normalize(T_h z_h) with T_h ANY fixed linear map (hand-designed OR learned), independent of t and of the content. Answer = argmax_j <z_H, C[j]>. No N x N score, no softmax over tokens, no KV cache. P3 (task). A chain k_0 -> k_1 -> ... -> k_n with DISTINCT keys; pair (k_p, k_(p+1)) is emitted as two ADJACENT tokens; pair ORDER is arbitrary; any number of zero fillers may appear between pairs; the query q_1 = C[k_0] is supplied externally; the answer is k_H. P4. H >= 2 and n >= 3 (so a distractor pair exists).

Claim. There EXISTS a legal dataset satisfying P1-P4 on which EVERY model satisfying P1-P2 outputs an answer different from k_H. Consequently zero role bits cannot realize retrieval that is correct for ALL legal inputs. The failure does NOT depend on the layer interface T_h, on the kernel coefficients, or on training: it follows from a single algebraic identity, A_1 = A_4, that holds for any fixed T_h.

Proof. It suffices to take H = 2; for H > 2 place the remaining pairs anywhere, the first two hops are unchanged. Choose five distinct ids k_0, k_1, k_2, w, k_3 and the pairs p_0 = (k_0,k_1), p_1 = (k_1,k_2), p_2 = (w,k_3). Emit them in the order [p_1, p_0, p_2] with no filler. The observed sequence is S = [k_1, k_2, k_0, k_1, w, k_3], L = 6. The chain is k_0 -> k_1 -> k_2, so the answer is k_2. This is a LEGAL instance of P3: the pair order is arbitrary by assumption. Step 1 (first hop is unambiguous). A_t = Phi_A C[x_(t-1)] for t = 0..5 equals 0, C[k_1], C[k_2], C[k_0], C[k_1], C[w] up to the fixed Phi_A. Because the keys are distinct and C has orthonormal rows, the matching signal m_1[t] = <q_1, A_t> is nonzero at exactly one position, t = 3, where x_2 = k_0. Hence z_1 = K_1(3) Phi_b C[x_3] = K_1(3) Phi_b C[k_1] and q_2 = normalize(T_1 z_1). CRUCIALLY q_2 is now FIXED and depends on neither the arrangement of the other pairs nor on the answer. Step 2 (the identity that forces failure). By P1 the predecessor symbol at t = 1 and at t = 4 is the SAME flat id k_1, therefore A_1 = A_4 = Phi_A C[k_1] exactly, for any Phi_A and any T_h. Write mu := <q_2, A_1> = <q_2, A_4>. This equality is the entire content of 'zero role bits'. Case (i) mu = 0. Then the only term carrying the answer code Phi_b C[k_2] -- namely b_1 = Phi_b C[x_1] = Phi_b C[k_2] -- is multiplied by m_2[1] = mu = 0 and vanishes from z_2. Since C's rows are orthogonal, no other position produces a C[k_2] component. z_2 is therefore independent of k_2, so replacing k_2 by any other id leaves the output unchanged while the answer changes. The model errs on one of the two. QED for case (i). Case (ii) mu != 0. The two matched positions contribute K_2(1) mu Phi_b C[k_2] and K_2(4) mu Phi_b C[w]. All other positions t in (2,3,5) have predecessor codes C[k_2], C[k_0], C[k_1]; if any of them happens to match q_2, its contribution is orthogonal to both C[k_2] and C[w] and can only make the answer LESS likely, never more. Let alpha = K_2(1)|mu| and beta = K_2(4)|mu|. Because every rho_hp < 1, K_h(t) is strictly increasing in t, and 4 > 1, so beta > alpha STRICTLY. Decoding gives logits_(k_2) proportional to alpha and logits_w proportional to beta; the maximum is therefore not k_2. The model errs. QED for case (ii). Both cases err, and the argument used no property of T_h, K_h or training beyond P2. Hence no model in the class is correct on all legal inputs. QED.

WITNESS (measured, not the proof): V=64, k_0=1, k_1=2, k_2=3, w=4, k_3=5, flat alphabet, identity interface: alpha=0.82605, beta=0.94950, beta>alpha, prediction=4 while answer=3. Under the 1-bit codebook partition beta=0 and prediction=3. With a LEARNED interface the flat model still predicts 4, and the measured A_1 == A_4 and b_1 != b_4 confirm the premise.

Boundary / scope. SCOPE. (i) This is a WORST-CASE / 'correct for all legal inputs' statement. Zero-bit models can be accurate on MANY inputs: measured flat accuracy is 0.7055 on the tight alternating layout, 0.8895 on gapped, 0.9711 on dense, 1.0000 on dense with a learned interface, because zero fillers implicitly suppress the spurious peak. We do NOT claim zero bits fail everywhere; we claim they cannot be exact everywhere, which is the relevant notion for a tightness theorem. (ii) P2 fixes the read-out to argmax over the orthogonal code components of z_H and the state to order 2. Architectures outside P2 -- learned input-dependent gates, extra recurrence, retaining intermediate layer outputs, or higher-order tensor states -- are NOT excluded; this mirrors R1 SCOPE (iii) and NT31's scope. (iii) P4 needs n >= 3 so a distractor pair exists; with a single pair there is nothing to confuse. (iv) Filler tokens are zero codes and are therefore identifiable by the model; this is NOT counted as role information (it distinguishes 'empty' from 'a symbol', not KEY from VALUE). (v) The alphabet convention of the 1-bit construction IS still an encoding-level inductive bias: KEY and VALUE symbols are different ids. What R3 establishes is that this is the MINIMUM such bias needed for uniform correctness, and R2 shows it need not be a run-time annotation, a position rule or a format prior. (vi) Synthetic content-chain task; no LM perplexity or generation claim. (vii) The proof is analytic. Numerical witnesses certify the premises and the conclusion, not the proof.

Use. Upgrades NT31 from a sufficiency result to a TIGHT characterization: order-2 state, exactly H layers, and exactly ONE role bit are each necessary and jointly sufficient. Converts v5.2's honest residual

into an optimality theorem, so the paper no longer needs to apologize for an alphabet convention. Also supplies the adversary construction that any future 'no role information' claim must defeat.

Witnesses (not the proof). Analytic proof above. Witnesses (NOT the proof): raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact, raw result files in the supplementary artifact. Code: the supplementary artifact

A.35 Lemma R4 - effective rank of the layer interface

Status. Supporting lemma; NOT a headline theorem or ledger item.

Premises Q1-Q4. Q1. Codebook C (V x V, V = 2 V_K) orthonormal, split C_key, C_val. Q2. Analytic optimum interface T* = C_key^T C_val. Q3. A dataset whose chains use at most m = n_pairs+1 DISTINCT keys in the VALUE half. Q4. Layer interface parameterized as T = U V^T / sqrt(r), U, V in R^(V x r), trained end-to-end; one learned output temperature tau > 0 (argmax invariant to tau).

Claim. (a) T* has AMBIENT rank exactly V_K as a V x V matrix. (b) On the data distribution Q3, restricting T* to the span of the m active VALUE codes has rank at most m; hence a rank-r interface with r >= m can represent the restriction exactly. (c) Parameter cost is 2 V r instead of V^2, and with the fast Walsh-Hadamard transform the codebook is never materialized. (d) Empirically r < m degrades and r >= 2m is exact.

Proof (a). C_key and C_val each have V_K orthonormal rows and are mutually orthogonal, so C_key^T C_val = sum_(j<V_K) e_j f_j^T with (e_j) and (f_j) orthonormal families; its rank is exactly V_K. QED.

Proof (b). Only VALUE codes C[V_K + k] with k in the active key set K (|K| = m) ever appear in z_h. Let P be the orthogonal projector onto span(C[V_K+k] : k in K). The action of T* that matters is T* P, whose image lies in span(C[k] : k in K), dimension m; hence rank(T* P) <= m. Write T* P = sum_(i=1)^(m) sigma_i u_i v_i^T and choose U = [u_1..u_m] diag(sqrt(sigma)) sqrt(sqrt(r)), V = [v_1..v_m] diag(sqrt(sigma)) sqrt(sqrt(r)) padded with zero columns to width r >= m; then U V^T / sqrt(r) = T* P. QED.

Proof (c). U and V hold 2 V r floats; the FWHT applies C in O(V log V) time and O(V) memory, so no V x V codebook is stored. At V = 2^18, V^2 = 6.87e10 entries (549.8 GB in fp64) while 2 V r with r = 8 is 4.19e6 parameters. QED.

Proof (d) is empirical, NOT analytic: measured 3-seed means are r=4 -> 0.9694, r=8 -> 0.9954, r=16/24/32/48/64 -> 1.0000, with m = 5. The observed threshold lies between m and 2m; we state it as a measurement, not as a proved sharp bound on the optimization path. NON-VACUITY: V=64, V_K=32, n_pairs=4, m=5, r=32 satisfies Q1-Q4 and attains 1.0000.

Boundary / scope. SCOPE. (i) (d) is a measurement about convergence of Adam on this task, not a theorem about the minimum trainable rank; the proved statement is (b), an existence result. (ii) m depends on the DATA distribution, not on V_K; for a corpus using the whole vocabulary, m approaches V_K and the ambient bound is recovered. (iii) The temperature tau is one extra scalar and does not affect argmax; it only makes the CE loss saturate (measured final loss 0.0014 versus ~3.0 without tau). (iv) T_h is a FIXED parameter: it does not vary with t or with the input, so Toeplitz / FFT-IIR duality are preserved (measured 2.89e-16). (v) Synthetic task; no LM claim.

Use. Removes the last hand-designed component of the architecture: the layer interface is learned end-to-end from a random initialization, yet five seeds reach exactly 1.0000 and the learned T aligns with T* (value->key block energy 0.9963). Together with R3, the ONLY remaining design choice is the 1-bit alphabet split, which R3 proves necessary. Gives a concrete parameter/FLOP budget at real vocabulary scale.

A.36 The final TSC-H layer specification

Appendix A.36 The final TSC-H layer (architecture specification)

Input. V = 2 V_K, C = Sylvester Hadamard(V), C C^T = I. Three symbol classes, distinguished ONLY by id (no run-time tag, no position rule):

KEY symbol k -> id k -> C[k] (first V_K rows)

VALUE symbol k -> id V_K + k -> C[V_K + k] (last V_K rows)

FILLER -> zero one-hot -> 0

This is the single role bit of R3. C is applied by FWHT, O(V log V) / O(V).

Per layer h = 1..H (K_h(t) = sum_(p=1..P) w_hp rho_hp^(L-1-t), 0<rho<1):

A_t = C[x_(t-1)]; b_t = C[x_t] (kernel-external)

S_h = sum_t K_h(t) A_t (x) b_t (order-2, O(1) state)

m_h[t] = <q_h, A_t> (content match)

z_h = sum_t K_h(t) m_h[t] b_t = S_h^T q_h

q_(h+1) = normalize(T_h z_h) (learned, FIXED in t)

T_h = U_h W_h^T / sqrt(r), r >= m = n_pairs+1 (R4)

Read-out: logits_j = tau <z_H, C[V_K + j]>, tau > 0 learned.

Invariants (all machine-checked on the FINAL learned-T construction):

Toeplitz deviation 0.0; FFT/IIR relative error 2.89e-16;

last-time deviation 1.45e-16; strict causality;

state = 2 P V floats per layer, independent of L;

no N x N score matrix, no softmax over tokens, no KV cache,

no input-dependent kernel, no external role tag, no position gate.

Cost: parameters = (H-1) * 2 V r + 1; codebook = 0 stored bytes (FWHT);

at V = 2^18 (Qwen3 scale) with r = 8: 4,194,305 parameters,

one FWHT 0.0119 s, versus 549.8 GB for a dense codebook.

A.1-A.30 Supporting statements (retained subset). Each entry states its premises, claim, proof, boundary and architectural use. These are a SUPPORTING evidence chain behind the four core results, NOT thirty independent discoveries. Statements belonging to a separate self-evolution engineering project were removed from this submission.

A.1 NT01 - Tight minimum tensor order for exact single-hop retrieval

Premises. Static LTI kernel K_n(d)>0 for every delay d>=0 (strictly positive); history contains n>=2 pairs (k_p,v_p) with independently assignable one-hot values; q=k_(j*); code rows satisfy <Code[a],Code[b]>=R*delta_(a,b); read-out is z_n = sum_(p,r) w_(p,n) q_r s_(p,r,n,L) with s_(p,r,n,t)=rho_(p,n)s_(p,r,n,t-1)+a_t[r]b_t[n].

Claim. (a) A vector-valued (order-1) state that is a fixed-dimensional linear summary of the history cannot implement the retrieval, for ANY read-out. (b) An order-2 tensor state attains it exactly, with zero interference and WITHOUT knowing the target position: z = R*K_n(L-tau_(j*))*v_(j*), hence argmax_n z_n = argmax_n v_(j*,n) because K_n>0. Therefore the minimum tensor order is exactly 2, and the bound is TIGHT.

(a) Let A=L(v_1..v_n) in R^d be the summary, L: R^(nd)->R^d linear and independent of q. rank(L)<=d<nd for n>=2, so dim ker L>=nd-d>0. Pick nonzero Delta-v in ker L supported on the target coordinate: two value assignments give identical (A,q) hence identical f(A,q) for any f, but require different answers. Contradiction.

(b) Unroll the recurrence: s_(p,r,n,L)=sum_tau rho_(p,n)^(L-1-tau) a_tau[r] b_tau[n]. Contract with q_r and w_(p,n): z_n = sum_tau (q . k_tau) K_n(L-1-tau) v_(tau,n) with K_n(d)=sum_p w_(p,n) rho_(p,n)^d. By code orthogonality q . k_tau = R delta_(tau,j*), so every non-target term vanishes exactly and z_n = R K_n(L-1-tau_(j*)) v_(j*,n). Since rho_(p,n)>0 and w>0 give K_n>0, the positive scalar cannot change the argmax over n. No position information is needed at read-out. Hence order 2 suffices.

Tightness: (a) forbids order 1 and (b) achieves order 2. QED

Boundary. Requires K_n(d)>0; a damped-oscillator kernel sin((tau+1)theta) changes sign and is NOT admissible. Requires one-hot/independently assignable values and an exactly orthogonal code; NT08 gives the robust version. Classification/argmax retrieval, not real-valued vector reconstruction.

Architectural use. Explains why the general real-pole dictionary beats the oscillator grid, and turns 'TSC works' into a tight optimality statement rather than an empirical accuracy. *Source:* Section 4 TSC; the anonymized reference implementation general second-order IIR dictionary

A.2 NT02 - Strict-positivity separation of the two pole families

Premises. Second-order IIR y_t = a y_(t-1) + b y_(t-2) + c u_t with characteristic roots either (i) two positive reals 0<r2<=r1<1, a=r1+r2, b=-r1*r2, or (ii) a conjugate pair rho*e^(+-i theta), 0<rho<1, 0<theta<pi, a=2 rho cos(theta), b=-rho^2.

Claim. Family (i) has h(tau)>0 for every tau>=0; family (ii) has h(tau)<=0 for some tau<=ceil(pi/theta). Hence the two impulse-response families are

DISJOINT, and membership is decided exactly by inspecting the sign sequence of h(0..L-1) in O(L) (zero misclassification in exact arithmetic). Moreover the map (r1,r2)->(a,b) is injective on the ordered real-root domain, so the real-pole parameterization has no redundant columns.

(i) h(tau)=sum_(j=0)^(tau) r1^(tau-j) r2^(j) = (r1^(tau+1)-r2^(tau+1))/(r1-r2) for r1!=r2, and (tau+1)r^(tau) for r1=r2=r. Every term is strictly positive, so h(tau)>0.

(ii) h(tau)=rho^tau sin((tau+1)theta)/sin(theta) up to a positive normalization. Let tau0=ceil(pi/theta). Then (tau0+1)theta >= pi+theta > pi and (tau0+1)theta <= pi+2theta < 3pi. If pi < (tau0+1)theta < 2pi the sine is negative; if it reaches [2pi,3pi) take tau0-1 instead, for which (tau0)theta in [pi, pi+theta) subset [pi,2pi), so sin<=0. In all cases some tau<=ceil(pi/theta) gives sin((tau+1)theta)<=0, hence h(tau)<=0 (rho^tau>0, sin(theta)>0). A strictly positive sequence cannot contain a non-positive term, so the families are disjoint.

Injectivity: (a,b) determines the monic polynomial z^2 - a z - b, whose roots are exactly (r1,r2); ordering r2<=r1 makes the pair unique. Real distinct roots require discriminant a^2+4b>0; both roots in (0,1) require 0<a<2 and -1<b<0. QED

Boundary. Root r1=r2 is a double real root (still strictly positive), handled by the limit formula. theta=pi is excluded by the architecture's theta clamp. Sign inspection needs exact arithmetic; quantization is covered by NT08.

Architectural use. Gives a provable, O(L)-checkable discriminant between the oscillator basis and the general second-order dictionary, and shows the latter has no parameter redundancy (unlike theta->0 collinearity, NT06). *Source:* the anonymized reference implementation _init_oscillator_grid / general second-order dictionary

A.3 NT03 - Provable separation of rho<1 multi-pole TSC from the linear-attention family

Premises. TSC read-out kernel K_n(d)=sum_(p=1)^(P) w_(p,n) rho_(p,n)^(d) with P>=1, 0<rho_(p,n)<1, w_(p,n)>0, all rho_(p,n) distinct. Linear-attention (single-pole, rho=1) kernel is the constant K(d)=1.

Claim. The span of (K_n : n) contains NO nonzero constant function, and the constant function is not in the closure of that span in l^infty(N). Hence the rho<1 multi-pole kernel family is STRICTLY SMALLER than the rho=1 single-pole family as a set of decay profiles, and the two state-update operators are not equal. In particular the TSC state is not the linear-attention state S=sum_tau k_tau v_tau^T once rho<1.

Any nonzero combination F(d)=sum_n c_n K_n(d) is a finite sum of exponentials with bases in (0,1). Let rho_max<1 be the largest base; then F(d)=Theta(rho_max^d) -> 0 as d->infinity. A constant function equal to 1 does not tend to 0, so no finite combination equals it. For the l^infty closure: ||F - 1||_infty >= sup_(d>=D) |1-F(d)| -> 1 for every F in the span, so the distance from the span to the constant is exactly 1 >0 and the constant is not a limit point either.

Consequently the rho=1 update s_t=s_(t-1)+u_t (constant kernel, state S=sum k_tau v_tau^T) and the rho<1 update s_t=rho s_(t-1)+u_t (decaying kernel) generate different linear operators on the same tensor signal; they cannot be identified by any change of basis that preserves the impulse response. The rho->1, P=1 limit recovers the isomorphism, which is therefore a boundary case, not the general case. QED

Boundary. This is a separation of KERNEL FAMILIES / state-update operators, not a claim that the full models have different Turing-complete expressivity, nor a claim of empirical superiority. It does not assert that TSC is outside every linear-attention VARIANT (e.g. data-dependent gates); it settles the specific rho<1 multi-pole vs rho=1 single-pole question.

Architectural use. Directly answers the reviewer's point 3: instead of leaving the TSC / linear-attention relation 'to the reader', we give a provable non-equivalence while still recording the rho->1 isomorphism. *Source:* Section 4 remark (rho->1 isomorphism) + core/theory/the anonymized reference implementation

A.4 NT04 - Deterministic zero-crosstalk codes for arbitrary V are 2-optimal

Premises. V>=1 arbitrary integer; code rows in R^R with zero crosstalk.

Claim. (a) Zero crosstalk among V nonzero rows forces R>=V. (b) R=2^ceil(log2 V), Code_V = first V rows of the Sylvester Hadamard H_R, satisfies Code_V Code_V^T = R I_V exactly and R<2V. Hence the construction attains approximation ratio strictly below 2 against the R>=V lower bound, and no Hadamard-prefix construction can do better than the next power of two.

(a) V mutually orthogonal nonzero vectors in R^R are linearly independent, so V<=R.

(b) Induction on the Sylvester recursion H_(2m)=[[H_m,H_m],[H_m,-H_m]] gives H_R H_R^T = R I_R; the leading V-by-V principal submatrix of that identity is exactly Code_V Code_V^T = R I_V. Since R=2^ceil(log2 V) < 2^(log2 V + 1) = 2V for V>1 (and R=V=1 for V=1), R/V < 2. Any Hadamard-prefix construction must use an order that is a power of two and at least V, so R is the minimum such order; the ratio is therefore optimal within this deterministic class. QED

Boundary. Existence + explicit construction + optimality within Hadamard prefixes. Does NOT prove deployed-kernel cheapness or that a non-Hadamard code cannot reach R=V (e.g. an orthonormal V-by-V basis would, but is not a deterministic +/-1 code).

Architectural use. Closes the V>>64 construction question with a clean 2-approximate bound. *Source:* CT5-L2 / core/theory/the anonymized reference implementation DETERMINISTIC_CODE

A.5 NT05 - State-vocabulary lower bound and end-to-end tightness

Premises. Static LTI tensor state s_(p,r,n) with P pole channels, R code rows, N value channels; exact zero-crosstalk retrieval of V token classes under NT01's premises.

Claim. Any such construction requires at least P*V*N scalar state registers, i.e. Omega(V*N). TSC with the NT04 code uses P*R*N < 2 P*V*N registers, so it attains the lower bound within a factor 2 END-TO-END: code dimension, state size and retrieval exactness are simultaneously tight.

Zero crosstalk requires R>=V mutually orthogonal code rows (NT04a). The state is indexed by (p,r,n); for each of the P*N pairs (p,n) the R code coordinates must be stored independently because the recurrence s_(p,r,n,t)=rho_(p,n)s_(p,r,n,t-1)+a_t[r]b_t[n] is diagonal in r and no linear combination of the r-coordinates is invariant under all one-hot inputs a_t=Code[x]. Formally, the observability matrix of the R parallel first-order systems has full column rank R, so a faithful state needs >=R registers per (p,n). Hence total >=P*R*N>=P*V*N. With R<2V from NT04b the construction uses <2P*V*N: ratio <2. NT01 supplies exactness, NT04 supplies the code, so all three bounds are met simultaneously. QED

Boundary. Lower bound is for faithful (injective-in-r) static LTI tensor states with a fixed code. Compressed or learned non-tensor states are outside the premise. The factor 2 is a guarantee, not optimality of the constant.

Architectural use. Turns the TSC construction into a tight state-complexity result instead of a single accuracy number on V=64. *Source:* Section 4 cost remark (56,320 B state) + NT01/NT04

A.6 NT06 - Quantitative rank-deficiency rate of near-zero frequency grids

Premises. Dictionary columns f_m(tau)=rho^tau sin((tau+1)theta_m), 0<rho<1, theta_1<theta_2<...<theta_M, Delta=theta_2-theta_1, Gram matrix G_(ij)=<f_i,f_j> over tau>=0.

Claim. cond(G) >= 2*sin^2(theta_1) / (Delta^2 * A(rho)), where A(rho)=sum_(tau>=0) rho^(2tau)(tau+1)^2 = (1+rho^2)/(1-rho^2)^3 < infinity. Thus a grid whose first spacing satisfies Delta << theta_1 forces cond(G) = Omega((theta_1/Delta)^2); a grid with Delta ~ theta_1/M and theta_1 = Theta(1) keeps the bound at O(M^2) instead of exponential.

Use the Rayleigh quotient with the vector e_1-e_2: sigma_min(G) <= (1/2)||f_1-f_2||^2. By |sin a - sin b| <= |a-b|, ||f_1-f_2||^2 = sum_tau rho^(2tau)[sin((tau+1)theta_1)-sin((tau+1)theta_2)]^2 <= Delta^2 sum_tau rho^(2tau)(tau+1)^2 = Delta^2 A(rho). The closed form follows from sum n^2 x^n = x(1+x)/(1-x)^3 with x=rho^2 and index shift. Also sigma_max(G) >= G_(11) = ||f_1||^2 >= rho^0 sin^2(theta_1) = sin^2(theta_1). Hence cond = sigma_max/sigma_min >= 2 sin^2(theta_1)/(Delta^2 A(rho)). For logfreq grids with theta_1 extremely small and Delta proportional to theta_1*(r-1), the two small quantities

nearly cancel in the RATIO but A(rho) explodes as rho->1; for linear grids theta_1=Theta(1) and Delta=Theta(1/M), giving cond >= Theta(1/M^2 * 1/A(rho)) while the measured value is O(1), i.e. the bound is not tight there but correctly predicts that near-zero theta_1 combined with tiny Delta destroys conditioning. QED

Boundary. A LOWER bound on cond (a badness certificate), not an exact value and not an upper bound. The linear-grid O(1) measurement is consistent with a loose lower bound; the theorem explains the FAILURE of near-zero grids, not the full success of linear ones. Condition numbers remain observations.

Architectural use. Upgrades the earlier 'direction only' asymptotic into a checkable rate, and explains why theta_min=pi/8192 collapses capacity while a linear grid restores it. *Source:* Prop8 / an internal technical report / the anonymized reference implementation init_mode

A.7 NT07 - Precision-length-decay trilemma with an interior optimal rho

Premises. First-order streaming state s_t=rho s_(t-1)+u_t, 0<rho<1, |u_t|<=U; each update is quantized with absolute step error <= delta, delta/U < 1; a distant contribution at delay L has amplitude rho^(L-1) U and must exceed c times the accumulated error floor (c>1 a fixed separation constant); c*delta/U < 1.

Claim. (a) The accumulated error is bounded by e_infty <= delta/(1-rho). (b) The resolvable length obeys L(rho) <= 1 + ln((1-rho)U/(c*delta)) / |ln rho|. (c) L(rho) is continuous on the open interval (0, 1-c*delta/U), tends to 0 at BOTH endpoints, and is positive inside; hence it attains an INTERIOR MAXIMUM at some rho* strictly between 0 and 1. Therefore rho->1 is NOT optimal: longer memory requires an interior decay, and rho=1 admits no resolvable length at all.

(a) The error obeys e_t <= rho e_(t-1) + delta; by induction e_t <= delta sum_(k=0)^(t-1) rho^k < delta/(1-rho).

(b) Resolvability requires rho^(L-1) U > c*delta/(1-rho), i.e. (L-1) ln rho > ln(c*delta/((1-rho)U)). Dividing by ln rho<0 reverses the inequality and gives L-1 < ln((1-rho)U/(c*delta))/|ln rho| =: g(rho).

(c) g requires (1-rho)U > c*delta, i.e. rho < 1 - c*delta/U. As rho -> (1-c*delta/U)^- the numerator -> 0^+ and the denominator is bounded away from 0, so g -> 0^+. As rho -> 0^+, |ln rho| -> infinity while the numerator tends to the finite positive ln(U/(c*delta)), so g -> 0^+. g is continuous and strictly positive somewhere inside (e.g. at rho=1/2 when delta/U is small), so by the extreme value theorem on a compact subinterval it attains a positive interior maximum. At rho=1 the error floor delta/(1-rho) is unbounded, so no L satisfies (b). QED

Boundary. CONDITIONAL on a uniform absolute quantization-error model (same premise as the perturbation bound). g is a SUFFICIENT resolvable-length bound, not an exact measured drop point; dynamic per-step scaling changes the effective delta, so O6's measured L=16385 is not derived from it. The existence of rho* is proved, its numerical value is an observation.

Architectural use. The flagship quantitative result: proves that the best decay is interior, explains rho=0.999 as a principled compromise, and turns 'rho is the only lever' from a measurement into a theorem with a rate. *Source:* O6 physics probes + Appendix A.7 perturbation bound + T12/T13

A.8 NT08 - Robust exact retrieval under bounded code perturbation

Premises. NT01's architecture, but the code satisfies Code Code^T = R I_V + E with ||E||_infinity <= epsilon; all kernels K_n(d) in [K_min,K_max] with K_min>0; values are one-hot with |v|<=1; at most L-1 non-target terms.

Claim. The retrieval remains exactly correct (argmax unchanged) whenever epsilon < R*K_min / ((L-1)*K_max). More generally the non-target interference is bounded by epsilon*(L-1)*K_max, so the target margin is at least R*K_min - epsilon*(L-1)*K_max.

Write q.k_tau = R delta_(tau,j*) + E_(q,tau). Then z_n = R K_n(L-1-tau_(j*)) v_(j*,n) + sum_(tau != j*) E_(q,tau) K_n(L-1-tau) v_(tau,n). The target coordinate of the one-hot value has amplitude >= R K_min. Each non-target term is bounded by epsilon*K_max*1 and there are at most L-1 of them, so the total interference on any coordinate is <= epsilon (L-1) K_max. If that is strictly less than R K_min, the target coordinate strictly dominates every other coordinate and argmax is unchanged. The stated inequality is exactly this condition. QED

Boundary. A SUFFICIENT condition, not necessary; assumes bounded kernels and one-hot values. It covers quantized codes, learned approximate codes and random orthogonal projections, explaining why O5's random-code arm also reached 1.0000 without any retraining.

Architectural use. Makes NT01 deployable: the deterministic Hadamard code is not a fragile requirement, only a bounded-crosstalk one. *Source:* Section 4 code remark + O5 random orthogonal code control

A.9 NT09 - Rounding-order crossover between the FFT and IIR paths

Premises. Standard floating-point model with unit roundoff eps; an L-point radix-2 FFT has log2(L) butterfly stages with relative error at most c1*eps per stage; the stable second-order IIR has steady-state relative error at most c2*eps/(1-rho), independent of L; c1,c2>0 fixed.

Claim. The IIR error bound is L-independent while the FFT bound grows as c1*eps*log2(L). They cross at L* = 2^(c2/(c1(1-rho))). For every L < L* the streaming path has the smaller worst-case bound; for rho close to 1, L* is astronomically large, so on all realizable sequence lengths the IIR bound is never worse.

FFT: after log2 L stages, (1+c1 eps)^(log2 L) - 1 <= c1 eps log2 L + O(eps^2). IIR: the perturbation recursion e_t <= rho e_(t-1) + c2 eps has the steady state e <= c2 eps/(1-rho), with no L dependence because the filter is BIBO stable and the bound is the l1 norm of its impulse response. Equating c1 eps log2 L = c2 eps/(1-rho) cancels eps and gives log2 L* = c2/(c1(1-rho)), i.e. L* = 2^(c2/(c1(1-rho))). Monotonicity of log2 L and constancy of the IIR bound give the two regimes. QED

Boundary. CONDITIONAL on the stated floating-point model and on WORST-CASE BOUNDS, not measured error. It does not contradict the withdrawal of unconditional fp32 bitwise duality (NT/Q02): equality is still only up to tolerance. It is a selection criterion between execution orders, not a speed claim.

Architectural use. Supplies the missing numerical-selection rule for the dual-form theorem: which path to trust at a given L and rho. *Source:* Thm1 remark + backend 16-op contract + Appendix A.7

A.10 NT10 - Per-channel time-invariance preserves the duality

Premises. TSC recurrence s_(p,r,n,t)=rho_(p,n) s_(p,r,n,t-1)+u_t[r,n] with rho_(p,n) depending on the pole channel p and value channel n but NOT on time t or code index r; coefficient tensor w_(p,n) likewise static.

Claim. Each (p,n) pair is an independent LTI system with transfer function H_(p,n)(z)=1/(1-rho_(p,n) z^(-1)). The full operator is BLOCK Toeplitz (Toeplitz in t for each channel, diagonal in (p,n)), so a single batched FFT diagonalizes it channel-wise and the streaming recurrence realizes the same operator. The state has P*R*N scalars, independent of L. Consequently a per-channel coefficient tensor of shape (M,D) is architecturally admissible without breaking any purity constraint.

For fixed (p,r,n), unrolling gives s_(p,r,n,t)=sum_tau rho_(p,n)^(tau) u_(t-tau)[r,n], whose impulse response depends only on the delay tau. Stacking the P*N independent channels gives a block-diagonal-in-(p,n) operator whose t-blocks are Toeplitz; by NT12 (applied per channel) the parameter must be constant in t, which it is. A Toeplitz operator is diagonalized by one DFT per channel, so one batched FFT suffices; the recurrence is the same rational function computed in the time domain. The state count is P*R*N with no L term. QED

Boundary. Requires rho and w independent of t. It does NOT permit input-dependent rho_t (that is the SFSC/LTV case, excluded by NT12/Q09). The constant P*R*N may be large (56,320 B observed); O(1) is asymptotic.

Architectural use. Upgrades the v4 finding (coef was artificially flattened from (32,2048) to 32 scalars) from a script simplification into a theorem about what the architecture permits. *Source:* the anonymized reference implementation coef (M,D) + T7 / Section 4

A.11 NT11 - Layer-count lower bound for multi-hop retrieval

Premises. A stack of L layers, each a polynomial map of total degree at most d in its inputs (a bilinear contraction has d=2); skip connections do

not raise degree; the target H-hop output contains H key-match factors and one value factor, i.e. total degree >= H+1 in the history variables.

Claim. Any such stack needs L >= ceil(log_d(H+1)) layers. For bilinear layers L >= ceil(log2(H+1)); in particular H=2 already requires L>=2, so a single contraction layer cannot implement two-hop retrieval. This is a NECESSARY condition; no sufficient construction is asserted.

The composition of polynomials of degrees d_1..d_L has degree at most the product d_1...d_L <= d^L (degree cannot increase under addition or skip connections). The H-hop target is a multilinear form of degree H+1 in (k,v): H factors of the form <phi(q),psi(k)> plus one value factor. A polynomial of degree < H+1 cannot equal a nonzero degree-(H+1) form on an open set. Hence d^L >= H+1, i.e. L >= log_d(H+1); taking ceil gives the integer bound. For d=2, H=2: log2(3)=1.58 -> L>=2. QED

Boundary. NECESSARY only. The degree bound is an upper bound on what L layers CAN reach; reaching it needs a construction (untested). Per-position nonlinearities of degree >2 change d. The measured H4_multihop=0.0322 is a finite-sample observation, not a theorem conclusion.

Architectural use. Gives the stacking roadmap a concrete quantitative target instead of 'try more layers', and explains the two-hop collapse structurally. *Source:* T15 / L7 / O5 H4_multihop; the anonymized reference implementation stacking

A.12 NT12 - Toeplitz iff time-invariant coefficients, for arbitrary order and channels

Premises. y_t = sum_(j=1..P) a_(j,t) y_(t-j) + c_t u_t with c_t != 0 for all t; P>=1 arbitrary; optionally with an independent copy for each of C channels (a_(j,t)^((c)), c_t^((c))).

Claim. The impulse response h(t,s) depends only on t-s IFF every coefficient a_(j,t) and c_t is constant in t. With C independent channels the full operator is block Toeplitz IFF this holds for every channel. Necessity fails at the first delay where c_t=0, so the theta-clamp / nonzero-c premise is essential, not decorative.

Necessity by induction on the delay k. k=0: h(s,s)=c_s must be s-independent, so c_t=c. Assume a_1..a_(k-1) are constant. At delay k, h(s+k,s) = a_(k,s+k) c + Phi(a_1..a_(k-1),c), where Phi is a polynomial with s-independent coefficients. Toeplitz forces h(s+k,s) independent of s, and c != 0 gives a_(k,t) constant. Induct to k=P. Sufficiency: constant coefficients give the LTI transfer function H(z)=c/(1-sum_j a_j z^(-j)), so h(t,s)=h(t-s). For C channels apply the same argument to each diagonal block; off-diagonal blocks are zero. If c_t=0 for some t, the k=1 step cannot isolate a_(1,t), and a counterexample with time-varying a_1 but Toeplitz h exists (the input never enters at that step). QED

Boundary. The equivalence concerns THIS convolution/FFT path; it does not assert nonexistence of other sub-quadratic algorithms. c_t != 0 is guaranteed in the spectral-convolution layer by sin(theta)>0 under the theta clamp; removing the clamp invalidates the necessity direction.

Architectural use. The precise equivalence-strengthening the reviewer asked to foreground; now covers arbitrary order and per-channel grids, and identifies exactly where the premise can break. *Source:* Thm4 + T11 + Q09/Q11; the anonymized reference implementation theta_clamp_floor

A.13 NT13 - Arbitrary-read-out impossibility with a sharp quantitative boundary

Premises. exactly four premises: (i) the history enters the final computation only through a fixed-dimensional LINEAR summary A=L(v_1..v_n) in R^d that is independent of q; (ii) q enters only at the current position; (iii) values are independently assignable; (iv) at least two distinguishable targets exist. Read-out f(A,q) is arbitrary (linear or nonlinear). Key linear independence is NOT required.

Claim. (a) Content addressing is impossible for any f. (b) Quantitatively dim ker L >= nd - d. (c) For n=1 the kernel can be zero, so premise (iv) is NECESSARY and cannot be dropped. (d) TSC escapes by changing the tensor order of the signal and the read-out, NOT by contradicting (a).

(a,b) rank(L)<=d, so dim ker L = nd - rank(L) >= nd-d > 0 for n>=2. Choose nonzero Delta-v in ker L supported on a coordinate of the target value (possible by (iii),(iv)). Two assignments v and v+Delta-v give the same A and the same q, hence the same f(A,q), but require different answers. Contradiction for every f. Linear read-out follows by subtracting two queries. (c) n=1 gives nd=d; a random square L is full rank almost surely, so ker=0 and (a) no longer follows -- the premise is sharp. (d) NT01 uses an order-2 tensor state, which is not a fixed vector summary, so (i) is false; the theorem is inapplicable, not violated. QED

Boundary. DOES NOT extend to deep stacks with per-position nonlinearities that mix A and q, to input-dependent kernels (NT12 instead), or to outer-product states (NT01). Corollary B1.1 ('requires a pairwise term') is an EXPLANATORY FRAMEWORK, NOT A THEOREM, and is not a proved sufficient or necessary characterization of all escape mechanisms.

Architectural use. The strongest boundary theorem, with an explicit four-premise contract that must appear in the abstract and conclusion to prevent misreading. *Source:* Thm5 + T10 + CT4; Appendix A.5

A.16 NT16 - Exact Turan identity and log-convexity of the real-pole diction

Premises. General second-order IIR y_t = a y_(t-1) + b y_(t-2) + u_t with characteristic roots 0 < r2 <= r1 < 1, so a = r1+r2, b = -r1*r2; impulse response h(tau) with h(0)=1, h(-1)=0.

Claim. (a) EXACT IDENTITY: h(tau)^2 - h(tau-1) h(tau+1) = (r1*r2)^tau > 0 for every tau >= 1, including the double-root case r1=r2=r where it becomes r^(2tau). Hence h is STRICTLY log-convex. (b) The conjugate-pole kernel h(tau) = rho^tau sin((tau+1)theta)/sin(theta) is NOT log-convex: it takes a negative value at some finite tau, whereas a log-convex sequence with h(0)>0 is strictly positive for all tau. (c) Therefore membership in the real-pole family is equivalent to log-convexity of the impulse response, a SUFFICIENT statistic that needs no root solving.

(a) Binet form: h(tau) = (r1^(tau+1) - r2^(tau+1))/(r1-r2) for r1>r2, and h(tau) = (tau+1) r^tau for r1=r2=r. For r1>r2 set A=r1^(tau), B=r2^(tau), p=r1, q=r2 and expand h(tau)^2 - h(tau-1)h(tau+1). Writing s_k = (p^(k+1)-q^(k+1))/(p-q) and using s_(n+1) = (p+q)s_n - pq s_(n-1), the Hankel determinant D_tau = s_tau^2 - s_(tau-1)s_(tau+1) obeys D_(tau+1) = pq D_tau with D_1 = pq, hence D_tau = (pq)^tau = (r1 r2)^tau > 0. The double-root case follows by continuity (or by the identity (tau+1)^2 - tau(tau+2) = 1), giving r^(2tau). Since D_tau>0 and h(0)=1>0, induction gives h(tau+1) > h(tau)^2/h(tau-1) > 0, so h is strictly positive and strictly log-convex.

(b) By NT02 there exists tau0 <= ceil(pi/theta) with sin((tau0+1)theta) <= 0, hence h(tau0) <= 0. A log-convex sequence with h(0)>0 satisfies h(tau)>0 for all tau by the same induction, contradiction.

(c) The two families are exhaustive for stable second-order roots (discriminant sign), and (a)/(b) separate them by a checkable inequality on the impulse response alone. QED

Boundary. Normalization h(0)=1 (c=1) is assumed; a general c>0 scales all h by c and leaves D_tau = c^2 (r1r2)^tau > 0, so log-convexity is invariant. theta=pi excluded by the architecture's clamp. The Turan identity is exact over R; finite precision is NT08/NT25.

Architectural use. Gives the STRUCTURAL reason the general second-order dictionary beats the damped-oscillator grid on content addressing (NT01 needs K_n>0), and turns NT02's O(L) sign scan into a log-convexity certificate. It explains why the 'oscillatory basis' was the wrong boundary. *Source:* the anonymized reference implementation general second-order IIR dictionary + NT02

A.17 NT17 - O(1) three-sample identifiability and pole-family discriminant

Premises. A stable second-order recursion with impulse response normalized to h(0)=1; the first three samples h(0),h(1),h(2) are observed exactly.

Claim. (a) The recursion coefficients are recovered exactly and uniquely: a = h(1)/h(0), b = (h(2) - a h(1))/h(0). (b) The 2x2 Hankel determinant det H_2 = h(0)h(2) - h(1)^2 = b != 0, so the minimal linear realization has rank exactly 2. (c) The discriminant Delta = a^2 + 4b computed from these three samples satisfies Delta = (r1-r2)^2 >= 0 for real poles and Delta = -4 rho^2 sin^2(theta) < 0 for conjugate poles, with equality Delta=0 iff the real roots coincide. Hence the pole family is decided in CONSTANT time from three samples, strictly stronger than NT02's O(L) sign scan and independent of L.

(a) From y_t = a y_(t-1) + b y_(t-2) + u_t with the unit impulse, h(0)=1,

h(1)=a h(0)=a, h(2)=a h(1)+b h(0)=a^2+b. Solving gives the stated inversion; uniqueness follows because (h(1),h(2)) determines (a,b) by a triangular system.

(b) det H_2 = h(0)(a^2+b) - a^2 = b. Real roots give b=-r1r2<0; conjugate roots give b=-rho^2<0; both nonzero, so rank(H_2)=2.

(c) The characteristic polynomial is z^2 - a z - b with roots (a +- sqrt(a^2+4b))/2. For real roots r1,r2 we have a=r1+r2 and b=-r1r2, so a^2+4b=(r1+r2)^2-4r1r2=(r1-r2)^2>=0. For a=2rho cos(theta), b=-rho^2 we get a^2+4b = 4 rho^2(cos^2 theta - 1) = -4 rho^2 sin^2 theta < 0 for theta in (0,pi). Equality Delta=0 forces r1=r2 (real) or sin theta=0 (excluded). Since (a,b) come from three samples, Delta does too, with no dependence on L. QED

Boundary. Requires exact h(0..2); with noise the discriminant is a finite-sample estimate (NT08 gives the robust version). Rank 2 is the MINIMAL LINEAR realization rank, not a claim about nonlinear realizations.

Architectural use. An O(1) architecture diagnostic: before training, three impulse samples tell whether a grid is real-pole or conjugate-pole and whether its roots coincide (collinearity risk, NT06). Useful for init_mode validation in the anonymized reference implementation. *Source:* the anonymized reference implementation _init_oscillator_grid / build_kernel

A.18 NT18 - Linear grids maximize the minimum spacing under a theta_min budget

Premises. M frequencies theta_1 < ... < theta_M in [theta_min, theta_max], theta_min > 0 fixed; let Delta_min = min_i (theta_(i+1)-theta_i). A grid is admissible if it respects the interval.

Claim. (a) Delta_min <= (theta_max - theta_min)/(M-1), with EQUALITY iff the grid is exactly equally spaced (linear). (b) Consequently, among all admissible grids, the linear grid uniquely maximizes the smallest pairwise gap, and therefore uniquely maximizes the NT06 conditioning lower bound among grids sharing (theta_min, theta_max, M). (c) Any logfreq grid whose local gaps near theta_min shrink geometrically has Delta_min <= theta_min(r-1), so its NT06 bound is strictly worse than linear's by a factor that grows as theta_min -> 0. A hybrid grid is a convex interpolation and is bounded between them.

(a) Summing the M-1 gaps gives theta_M - theta_1 <= theta_max - theta_min. Hence (M-1) Delta_min <= sum gaps <= theta_max - theta_min, giving the bound. Equality requires every gap equal AND theta_1=theta_min, theta_M=theta_max, i.e. the linear grid.

(b) NT06 gives cond >= 2 sin^2(theta_1)/(Delta^2 A(rho)); with theta_1 and A(rho) fixed by the budget, maximizing the bound means maximizing Delta_min, which (a) shows is the linear grid, uniquely.

(c) For a logfreq grid theta_(i+1)=r theta_i, the smallest gap is theta_1(r-1) <= theta_min(r-1) (equality at the first gap). A hybrid grid is a mixture theta_i = alpha*linear_i + (1-alpha)*logfreq_i; its minimum gap is at least alpha*Delta_linear + (1-alpha)*Delta_logfreq and at most Delta_linear, so its conditioning bound lies between the two. QED

Boundary. Optimizes the MINIMUM gap / the NT06 LOWER bound, not the exact condition number (which is a measured value). It does not claim linear is best for every task; NT06's bound is loose at large theta_1. Theta ranges and M are fixed by the budget.

Architectural use. Turns 'linear30 was best in a 3-boundary sweep' from a measurement into a uniqueness theorem, and explains quantitatively why hybrid sits between legacy and linear (an internal technical report). *Source:* an internal technical report + the anonymized reference implementation init_mode

A.19 NT19 - The two-tap state is a MINIMAL sufficient statistic (tight byte bound)

Premises. Stable second-order IIR with distinct OR repeated roots, per basis function m; D channels; M basis functions; streaming state stores the previous two outputs per (m,d).

Claim. (a) No deterministic state of fewer than 2 scalars per (m,d) can reproduce the same output sequence for all inputs. (b) Hence 2*M*D scalar registers are NECESSARY, not just sufficient, and the architecture's state expression 4(2MD + (K-1)D) bytes is tight up to the gate registers. (c) The O(1) state property (Thm3) is therefore OPTIMAL within causal LTI realizations: the constant cannot be reduced without changing the kernel family.

(a) Suppose a 1-scalar causal linear state s_t = lambda s_(t-1)+u_t realizes the operator. Its impulse response is h(tau)=lambda^tau, so h(1)^2 = h(0)h(2). But NT17 gives h(0)h(2)-h(1)^2 = -b != 0 for BOTH root families (b=-r1r2 or b=-rho^2). Contradiction. A nonlinear 1-scalar state cannot realize a linear operator whose Hankel rank is 2 (rank is realization-independent), so 2 is a lower bound for any CAUSAL LINEAR state realization. (Nonlinear states of unbounded precision are outside this premise; the architecture uses fixed-width fp32 registers.) The recurrence itself achieves 2, so the minimum is exactly 2.

(b) M basis functions and D channels are independent (block diagonal in (m,d)), so the minima add: 2MD. Adding the K-1 gate scalars per channel gives the source expression; the L-independent part is exactly the necessary part.

(c) Any causal LTI realization of a degree-2 rational transfer function has McMillan degree 2; by (a) this is a hard lower bound, so no re-parameterization reduces the constant. QED

Boundary. Minimality is within CAUSAL LINEAR state realizations of the SAME kernel (the family used by the anonymized reference implementation); unbounded-precision nonlinear registers are outside the premise. Changing the kernel family (e.g. first-order rho-only) changes the premise. Gate registers are architecture-specific overhead, not part of the state bound.

Architectural use. Answers the reviewer's 'state constant is large' concern with a tight lower bound: the 56,320 B TSC state is a DIFFERENT (tensor) construction, while the original mixer's 512 B cannot be shrunk without losing the second-order kernel. *Source:* the anonymized reference implementation streaming_state_bytes + Thm3 + NT17

A.25 NT25 - Tightness of the quantization-resolvable radius

Premises. CONDITIONAL on a uniform-step absolute quantizer with step s, input bound |u_t| <= U, r = s/(2U) < 1, first-order state s_t = rho s_(t-1) + u_t with 0<rho<1; a contribution at delay tau is resolvable iff rho^tau U > s/2.

Claim. (a) The resolvable radius is exactly tau* = ln(r)/|ln rho| (floor), and it is TIGHT: there exists an input with a single unit pulse such that the contribution at tau*+1 is exactly at the quantization threshold and cannot be resolved, while all delays <= tau* are resolvable. (b) tau* = Theta((m+1)/|ln rho|) for m-bit mantissas, so the asymptotic rate is OPTIMAL up to a constant: no deterministic rescaling improves the EXPONENT. (c) For rho = 1-epsilon, tau* ~ (m+1) ln 2 / epsilon, giving the observed order of magnitude of the bf16/int8 long-length drop.

(a) rho^tau U > s/2 iff tau ln rho > ln(s/(2U)) = ln r, and since ln rho<0 this is tau < ln r / ln rho = ln(r)/|ln rho|. The largest integer below it is tau*. Choose the input with u_0=U and all other u_t=0: the state at delay tau is exactly rho^tau U, so at tau*+1 it equals or falls below s/2 by construction, hence is not resolvable; for every tau<=tau* it strictly exceeds s/2. No scaling of U or s changes r's logarithm by more than an additive constant.

(b) With s/U ~ 2^(-(m+1)), ln(1/r) = (m+1) ln 2 + O(1), so tau* = ((m+1) ln 2 + O(1))/|ln rho|. The dependence on |ln rho| in the DENOMINATOR is forced by the exponential decay law, so no deterministic rescaling alters the exponent; it can only change the O(1) numerator.

(c) |ln(1-epsilon)| = epsilon + O(epsilon^2), giving tau* ~ (m+1) ln 2 / epsilon. QED

Boundary. CONDITIONAL on a uniform absolute quantizer; the physics probe uses dynamic per-step scaling, so MEASURED drop points (L=2049/4097) are NOT derived from this theorem -- only the order of magnitude is. Tightness is within this model, not for all quantization schemes.

Architectural use. Upgrades the earlier 'conservative sufficient radius' into a TIGHT radius with an explicit precision-vs-decay rate, and explains NT07's interior optimum quantitatively. *Source:* Appendix A.7 perturbation bound + O6 physics probes + NT07

A.26 NT26 - Causal optimality of per-step peak calibration

Premises. CONDITIONAL. Streaming state s_t, causal information set at step t contains only I_t = (|u_s| <= U, s_(t-1), rho), i.e. no future peak is

known. A quantizer chooses a scale Delta_t >= |s_t| to avoid overflow and incurs absolute error delta_t proportional to Delta_t. A 'fixed global scale' uses Delta_t = Delta for all t.

Claim. (a) The per-step peak rule Delta_t = max(|s_t|, Delta_min) is the POINTWISE MINIMAL admissible scale given I_t, hence minimizes delta_t at EVERY step. (b) Any fixed global scale must satisfy Delta >= sup_(t<=L)|s_t|, and since sup >= max_t|s_t| pointwise, delta_fixed >= max_t delta_peak(t), with strict inequality whenever the peak varies. (c) Therefore fixed-scale quantization is never better and is strictly worse on any sequence with a non-constant state envelope; the improvement is CAUSAL (needs no future information).

(a) Admissibility requires Delta_t >= |s_t|; the smallest such value is |s_t| (or Delta_min for the zero state), which uses only s_(t-1), u_t and rho -- all in I_t. Minimality is pointwise, so delta_t = c Delta_t is minimized pointwise.

(b) A fixed Delta must cover every t<=L, so Delta >= sup_t |s_t| >= |s_t| = Delta_peak(t) for each t; hence delta_fixed >= delta_peak(t) for all t, i.e. >= max_t delta_peak(t). If the envelope is non-constant the inequality is strict at the step where |s_t| < sup.

(c) Combining (a) and (b) gives the partial order, and (a) shows the rule is implementable causally. QED

Boundary. CONDITIONAL on scale-proportional absolute error and the stated causal information set. It proves a PARTIAL ORDER of error bounds, NOT that dynamic scaling always wins empirically (O6's fp16 non-monotonicity is a separate genuine numerical effect). It does not contradict the withdrawal of 'dynamic scaling causes spurious artifacts' -- those are different claims.

Architectural use. Upgrades the previously DEMOTED one-line partial order (T14) into a causal-optimality theorem, and is exactly the standard that separates 'too ordinary to write' from 'non-trivial enough'. *Source:* O6 fixed-scale controls + T14 + docs physics frontier

A.27 NT27 - Phase transition of the optimal decay across precision depths

Premises. CONDITIONAL. NT07's resolvable-length bound g(rho) = ln((1-rho)U/(c delta)) / |ln rho| on rho in (0, hi), hi = 1 - c delta/U; delta/U = 2^(-(m+1)) for mantissa depth m; c>1 fixed; delta/U sufficiently small.

Claim. (a) g is strictly increasing near rho -> 0 and strictly decreasing near hi, so for small delta/U it has a SINGLE interior maximum rho*(m). (b) rho*(m) solves the stationarity equation (1-rho) ln(1/rho) ln((1-rho)U/(c delta)) = rho ln(1/rho) + ... equivalently the balance (1-rho) ln((1-rho)U/(c delta)) = rho ln(1/rho); asymptotically rho*(m) = 1 - Theta(2^(-m)). (c) Therefore DIFFERENT precision depths have DIFFERENT optimal decays: a single rho tuned for fp32 is provably suboptimal for int8. rho -> 1 is never optimal (NT07).

Write N(rho)=ln((1-rho)U/(c delta)) > 0 and D(rho)=ln(1/rho)>0, g=N/D. Then g' = (N' D - N D')/D^2 with N' = -1/(1-rho) < 0 and D' = -1/rho < 0, so sign(g') = sign(-D/(1-rho) + N/rho). As rho -> 0, D = ln(1/rho) -> +infinity but N -> ln(U/(c delta)) is finite and positive; the term N/rho ~ ln(U/(c delta))/rho grows as 1/rho and therefore dominates -D/(1-rho) ~ -ln(1/rho). Hence g' > 0 near 0. As rho -> hi, N -> 0+ while D stays positive and finite; then N/rho -> 0 and -D/(1-rho) < 0 dominates, so g' < 0 near hi. By continuity g' has a zero in (0,hi); for small delta/U the ratio N/D is unimodal (one sign change), giving a UNIQUE interior maximum rho*. Setting g'=0 gives N/rho = D/(1-rho), i.e. (1-rho) N = rho D, the stated balance. Now put rho = 1-epsilon with epsilon small and delta/U = 2^(-(m+1)): then D = ln(1/(1-epsilon)) = epsilon + O(epsilon^2) and N = ln(epsilon * 2^(m+1)/c) = (m+1) ln 2 + ln epsilon - ln c. The balance (1-rho)N = rho D becomes epsilon * N = (1-epsilon) * (epsilon + O(epsilon^2)), so for epsilon>0 we need N = 1 + O(epsilon). Substituting N gives ln epsilon = 1 - (m+1) ln 2 + ln c + O(epsilon), hence epsilon = c*e*2^(-(m+1)) (1+o(1)) = Theta(2^(-m)) and rho* = 1 - Theta(2^(-m)). This matches the NT07 finite scans (m=8 -> 1-rho* ~ 1e-2, m=11 -> ~1e-3, m=24 -> ~3e-8). Since rho* depends on m, a single rho cannot be optimal for two different mantissa depths. QED

Boundary. CONDITIONAL on NT07's uniform-error model; uniqueness of the stationary point is asserted for SMALL delta/U (the regime of interest); it is not a global proof for all parameters. The asymptotic rho*=1-Theta(2^(-m)) is analytic, and is CROSS-CHECKED against NT07's finite scans but not derived from them. Measured O6 rho=0.999 is an observation.

Architectural use. Explains why the int8 / bf16 / fp32 frontiers must be tuned separately and gives the precision-dependent decay law; strengthens NT07 from 'an interior optimum exists' to 'how it moves with m'. *Source:* NT07 + O6 raw result files in the supplementary artifact

A.28 NT28 - Layer x tensor-order incompatibility inequality for H-hop retr

Premises. A stack of L layers; each layer's output is a polynomial of total degree at most the tensor order o of its state contraction (order-2 outer product gives o=2; bilinear contraction degree 2). Skip connections do not raise degree. The H-hop target contains H key-match factors plus one value factor, total degree H+1.

Claim. L * ln(o) >= ln(H+1), equivalently o^L >= H+1. For o=2 this recovers NT11 (L >= log2(H+1)); for L=1 it forces o >= H+1, so a SINGLE layer needs tensor order at least H+1. CRITICALLY: increasing the number of pole channels P changes the KERNEL SHAPE but not the tensor order o, so P cannot compensate for too few layers or too low an order. The two resources (depth, order) are incomparable and must both satisfy the product bound.

Composition of polynomials of degrees d_1..d_L has degree at most prod d_i <= o^L; skip connections (addition) do not increase degree. The H-hop target is a multilinear form of degree H+1 (H factors <phi(q),psi(k)> plus one value). A polynomial of degree < H+1 cannot equal a nonzero degree-(H+1) form, so o^L >= H+1, giving the inequality. Setting o=2 gives NT11; setting L=1 gives o>=H+1. The pole count P enters only through K_n(d)=sum_p w_(p,n) rho_(p,n)^d, a SCALAR multiplier of each delay; it does not add a new tensor index or raise the polynomial degree in the history variables. Hence P is orthogonal to the bound. QED

Boundary. NECESSARY condition only; no sufficient construction. The degree bound is an upper bound on what L layers of order o can reach. Per-position nonlinearities of degree > o change the premise. The measured H4_multihop=0.0322 is a finite-sample observation.

Architectural use. Unifies NT01 (order 2 for 1 hop) and NT11 (layers for H hops) into one inequality, and PROVES that the 'add more poles' fix cannot work -- closing a whole search direction that was tried implicitly. *Source:* NT01 + NT11 + O5 H4_multihop

A.29 NT29 - Topological necessity of a discontinuous hard pointer

Premises. Exact retrieval map R(q) = v_(j*(q)) with j*(q) = argmax_j <phi(q), k_j>, where >=2 distinct keys k_1,k_2 have distinct values v_1 != v_2 and the key space is a continuum containing the bisector hyperplane between them. The state is constant-size (no per-token cache).

Claim. (a) R is DISCONTINUOUS on the bisector, hence not differentiable there; hard pointer selection is therefore a NECESSARY feature of exact retrieval, not an implementation artifact. (b) Any CONTINUOUS surrogate Rtilde that agrees with R off an epsilon-neighborhood of the bisector has a NONZERO mis-selection probability bounded below by the measure of that neighborhood under any query distribution with positive density there. (c) Softening cannot simultaneously be continuous AND exact; this is an INDEPENDENT obstruction from NT14(iii)'s kernel-collision argument (which concerns the aggregate read-out, not the selector).

(a) Let q_1,q_2 be points on opposite sides of the bisector B = (q : <phi(q),k_1> = <phi(q),k_2>) with <phi(q_1),k_1> > <phi(q_1),k_2> and the reverse at q_2. Then R(q_1)=v_1 and R(q_2)=v_2. Take a path q(s) from q_1 to q_2 crossing B at s0; the left limit of R is v_1 and the right limit is v_2 != v_1, so R is discontinuous at s0 and has no derivative there.

(b) A continuous Rtilde must pass from v_1 to v_2 continuously, so on the epsilon-neighborhood N of B it takes values strictly between them; no point of N is mapped to the exact correct value v_1 or v_2. Hence P[Rtilde(q) != R(q)] >= P[q in N] > 0 for any query law with positive density on N.

(c) NT14(iii) shows a differentiable SOFT AGGREGATE over all slots has a kernel collision; (a)-(b) show the SELECTOR itself cannot be continuous. They are different maps with different proofs, so the two obstructions are independent. QED

Boundary. Requires a continuum query space and >=2 distinguishable

keys; a finite/discrete query alphabet could in principle use a lookup table (but that is itself a discontinuous map on the embedding space). Does not forbid approximate retrieval (NT08).

Architectural use. Gives the L0/L1/L2 three-tier sidecar its theoretical justification: the L2 top-k POINTER must be hard/non-differentiable, which is why it sits outside the differentiable the spectral-convolution layer mixer and cannot be folded into end-to-end training. *Source:* docs L0/L1/L2 tier design + the anonymized reference implementation topk + NT14

A.30 NT30 - End-to-end tightness chain: none of the five bounds can be improved

Premises. Exact H-hop retrieval over V token classes, N value dimensions, M second-order basis functions, in a static-LTI constant-state architecture with no N x N scoring and no growing KV cache; NT01, NT04, NT05, NT19, NT28 and NT13 all hold.

Claim. Any such architecture simultaneously requires (i) tensor order o >= H+1 for L=1, or o^L >= H+1 in general (NT28); (ii) code dimension R >= V (NT04); (iii) state >= 2*M*R*N scalars, with R < 2V attainable (NT05, NT19); (iv) a strictly positive kernel, hence real-pole roots (NT16/NT01). TSC at H=1 attains ALL FOUR within constant factors: order 2, R<2V, 2M*R*N<4M*V*N, real poles. Moreover the chain is JOINTLY TIGHT: improving any one bound alone cannot lower the product, because each is forced by a different independent theorem.

Chain the results: NT13 forbids order 1; NT01 achieves order 2 (tight). NT04(a) gives R>=V and NT04(b) gives R<2V (2-optimal, tight within the class). NT19 gives 2M scalars per code coordinate per value channel as a MINIMAL sufficient statistic, so with R code rows and N value channels the state is >= 2MRN; TSC uses exactly P*R*N with P=M poles, i.e. M*R*N first-order registers, and the second-order version 2MRN is attained by the recurrence (tight). NT16 forces strict positivity, hence real poles by NT02. For H hops NT28 forces o^L >= H+1, which no change in P, R, M or N can relax. Each inequality comes from a DIFFERENT proof (rank-nullity / orthogonality / Hankel rank / Turan / degree composition), so relaxing one would contradict its own theorem without affecting the others: the product bound is jointly tight. TSC at H=1 meets all four simultaneously, so the bounds are not merely lower bounds but ACHIEVED ones. QED

Boundary. Tightness is up to constant factors (the factor 2 in R<2V, and the pole count P=M convention). It does NOT assert that no architecturally DIFFERENT construction (e.g. a non-LTI state) could beat the product; that would leave the premise. Neither LM parity nor throughput is implied.

Architectural use. The paper's single strongest summary statement: instead of one accuracy number, an end-to-end achieved tight bound. It also PROVES the project cannot be improved by 'add more poles / bigger grid / better code' alone -- only by leaving the premise (NT28). *Source:* NT01/NT04/NT05/NT16/NT19/NT28 consolidated

Appendix B Conjectures and Open Questions

Explicitly unverified. None is a premise of any theorem or a claimed result. NT04 closes C2; NT11 gives C1/C3 a necessary condition but no construction.

· **C1.** Stacking TSC layers or inter-layer recurrence restores multi-hop retrieval. NT11 gives the layer lower bound; no stacking experiment exists.

· **C2.** CLOSED by NT04: a deterministic orthogonal code exists for every V with R < 2V.

· **C3.** NT13 extends to deep stacks with per-position nonlinearities. Cross-terms between A and q leave the fixed-summary premise; unproved.

· **C4.** Freezing rho = 0.999 from the start of training remains stable on >=10M-token language modeling. Verified only on a synthetic task.

· **C5.** Layered quantization or residual accumulation pushes the int8 long-length boundary further. NT07 bounds the trade-off; untested schemes remain.

· **C6.** On equal compute the periodic and content-addressing axes are complementary. The periodic-axis gain was NOT replicated (O4).

· **C7.** CLOSED IN PART by NT03: the rho<1 multi-pole kernel is provably not the linear-attention constant kernel. Full-model separation from all gated variants remains open.

· **C8.** The decay spectrum gives an inductive bias on oscillatory/damped physical signals that attention lacks. No measured comparison exists.

Appendix C Falsified Statements

· **F1.** Increasing rho_max extends the effective memory window. Rejected by measurement.

· **F2.** State-feedback rho_t/theta_t modulation (SFSC) preserves FFT duality. Rejected: LTV is non-Toeplitz (NT12); FFT-proxy relative error 0.9720.

· **F3.** Phase-resonance slots exceed equal-parameter attention. Rejected (O2).

· **F4.** Content addressing and FFT/IIR duality are mutually exclusive. WITHDRAWN via NT01/NT10; this withdrawal is a core contribution.

· **F5.** An oscillatory basis itself provides the inductive advantage. Rejected: a random orthogonal basis reached 1.0000; NT02/NT08 explain why.

· **F6.** Dynamic quantization scaling causes spurious non-monotonicity. Rejected by same-sample prefix controls.

· **F7.** Shannon capacity limits long-length retrieval here. Rejected: 176x headroom; NT07 identifies the decay/precision trade-off instead.

· **F8.** A monotone verifier score implies monotone self-improvement (former T16). WITHDRAWN (W7); trivial scalar induction and inconsistent with the multi-metric paired-bootstrap gate. Replaced by NT15.

Appendix D Observation Provenance

All values were computed from raw result files accompanying the anonymized supplementary artifact; none was transcribed from a secondary summary.

Obs.	Config / seeds	Key values	File
O1	1.7B transfer, 1 run	0.2484 / 0.2476 / 0.5430	transfer study
O2	single-hop, 3 seeds	0.8672 vs 1.0000	phase-resonance probe
O3	470K, 3.07M tokens	10.817 / 11.281 / 10.217	native pilot
O4	7 grids, 1 run	0.1094-0.1354 vs 0.1575	periodic-axis probe
O5	V=64, 5 seeds, 512/seed	1.0000 / 0.9987 / p=0.4998	TSC probe
O6	5 seeds, 4 precisions	L=16385, 43/43 checks	physics probes

Measured configuration details. O5: W1 has 10264 parameters, W4_half 4920, M5 2792; the stored reference record has 10240. **Fisher p = 0.4998 means no significant difference was detected, NOT equivalence.** The paired bootstrap on the general second-order dictionary gave mean difference 0.1901 with 95% CI [0.1582, 0.2220] over 3 seeds (sample-internal variance only). O6: frozen-rho retraining held-out means [1.0, 1.0, 1.0, 1.0], between-seed SDs [0.0, 0.0, 0.0, 0.0]; L = 16,385 accuracy mean 1.0000, min 1.0000; per-token cost at L = 4097 was 35.25 us with a 56320 B state. Conservation guards reported FFT/IIR relative errors of order 1e-16.

Appendix E Reproducibility and Tool Use

Every theorem is derived analytically in Appendix A and requires no execution. Every observation comes from a deterministic script with recorded seeds, precision and sample counts. The supplementary artifact contains raw result files, gradient finite-difference checks, conservation assertions, the 49-item audit (49/49 PASS) and the theorem ledger. Gradient checks passed 8/8, 10/10, 28/28 and 44/44; the physical consistency checker passed 43/43. No training run is used to support any theorem.

Generative AI tools were used for implementation assistance, experiment scripting and language editing. They did not originate the research questions, conjectures or conclusions; no experimental data, citation or result was generated by such a tool. The human author verified every numerical claim against raw files and accepts full responsibility.

Appendix F Withdrawn Claims Register

Earlier working-version statements, formally withdrawn so the revision is auditable. Withdrawals are contributions, not hidden errata.

Withdrawn statement	Reason
Unconditional fp32 bitwise duality	Different rounding orders; real-field only (NT09)
Content addressing <-> duality exclusive	WITHDRAWN, core contribution: NT01/NT10 counterexample
Periodic-axis superiority over attention	Not replicated in-repository (O4)
Equivalent 4-7B parameter claim	No supporting measurement
NT13 holds for any read-out unrestricted	Exactly four premises required; stated in abstract
Dynamic scaling causes spurious effects	Same-sample prefix controls (O6)
W7: monotone score => self-improvement (T16)	Trivial induction and inconsistent with the multi-metric gate
F8: scalar monotone score as gate	The gate is the unique multi-metric AND rule
our earlier mutual-exclusivity claim: content addressing needs input-dependent coefficients	WITHDRAWN, second core contribution: the layer-tightness theorem attains H-hop retrieval with STRICTLY static kernels
the withdrawn positional format prior: absolute-position parity is a general role basis	WITHDRAWN: hidden format prior; 1.0000 -> 0.3438+/-0.0314 under arbitrary spacing. v6 CONVERGENCE COMPLETE: parity -> 1-bit tag -> endogenous codebook -> learned interface; R3 proves the single remaining bit NECESSARY and MINIMAL.

Appendix G Statement Classification

The submission's contribution is one main theorem (A), two tight structural bounds (T1, T2), one constructive counterexample, and four falsifiable witness groups. Everything else is classified, not promoted. Supporting lemmas and corollaries (R1-R4) have complete proofs in Appendix A; standard facts are propositions; falsified and withdrawn statements are recorded in Appendices C and F as a SOUNDNESS practice, not as an originality claim; empirical claims about scale, perplexity or superiority are explicitly marked as unestablished. A verified count of statements is neither a contribution nor a substitute for one.

The full engineering ledger, including the removed self-evolution project items, is available in the public technical report and is NOT part of this submission.